Contents

1 Introduction

1.1 Background and Significance

Global CO_2 concentrations in the atmosphere are among the most important environmental indicators of today, and this measure has now been the main indicator of greenhouse gases trapped in the atmosphere of the planet. Mauna Loa Observatory in Hawaii is the longest time series of the measurement of CO_2 content of the atmosphere since 1958, and has formed the subject of much study in climate science. The trend in concentrations of CO_2 is crucial to climate policy making, the evaluation of environmental impacts, as well as global warming prevention plans.

The time series data of CO_2 has somewhat different properties like intense trend (high global emissions) and seasonal signals (Northern hemisphere vegetation cycle) as well as possible volatility bandwidth in recent decades, which makes the analysis difficult. The above characteristics qualify the data on CO_2 concentration to be a superb choice to implement complex time series modeling with ARIMA, SARIMA, and GARCH techniques.

1.2 Research Objectives

In the present report, a complete time series analysis of CO_2 monthly global data concentration of the Mauna Loa Observatory is performed, and the period of analysis is April 2015 to April 2025.

Research Question: How effectively can ARIMA, SARIMA, and GARCH models forecast monthly CO_2 concentration levels at Mauna Loa for the next 10 months?

The main goals of the same research are:

- Carry out proper descriptive analysis to define the time series behavior of the CO_2 concentration
- Exploit temporal dependencies by discerning and fitting proper ARIMA patterns to the data
- Model Seasonality in atmospheric CO_2 cycles using SARIMA methodology
- Use GARCH models to look at volatility clustering and conditional heteroscedasticity
- Provide precise predictions on the future of the next 10 months through the best match of the model
- Test and compare model performance by doing thorough diagnostic testing

1.3 Data and Computational Tools

The data pool contains monthly CO_2 readings recorded in the Mauna Loa observatory between April 2015 and April 2025, which offers 121 points with a 10-year time gap. This data is comprised of both raw measurements and seasonally adjusted values and reflects not only long-term trends, but also cyclical changes in atmospheric CO_2. During this period, some major global events have occurred, such as the implementation of the Paris Climate Accord and the effects of the COVID-19 pandemic on the level of emissions.

R programming language and packages for time series analysis are employed in analysis: `forecast` and `rugarch` packages provide implementation of ARIMA/SARIMA and GARCH models, respectively, `TSA` package allows to test diagnostics, and visualization tools are used to perform a detailed graphical interpretation. RStudio is the preinstalled development environment guaranteeing the reproducible research practice.

1.4 Analytical Approach

The methodology is structured in the form of a stepwise process that starts with exploratory data analysis in order to determine the main features of the CO_2 time series. Testing the stationarity is done with the use of a series of formal tests (ADF, KPSS, Phillips-Perron) and supplemented with visual diagnostics. ACF, AIC, PACF, EACF, and the information criteria (AIC, BIC) are used to specify the model by finding the candidate models.

They use three modeling methods, namely: (1) ARIMA models of a non-seasonal dependence, (2) SARIMA models having seasonal components, and (3) GARCH models to analyze volatility. The diagnostic test applied to each model is intensive, involving residual analysis, the test of normality, and prediction accuracy. The best model regarding statistical selection as well as applicable forecast performance, is chosen.

2 Preliminary Exploration

2.1 Data Summary

```
##    year month decimal.date average deseasonalized ndays sdev  unc
## 1  2015     4     2015.292  403.35         400.46    26 0.86 0.32
## 2  2015     5     2015.375  404.15         400.70    30 0.32 0.11
## 3  2015     6     2015.458  402.97         400.65    29 0.47 0.17
## 4  2015     7     2015.542  401.46         401.11    24 0.57 0.22
## 5  2015     8     2015.625  399.11         401.04    28 0.74 0.27
## 6  2015     9     2015.708  397.82         401.43    25 0.32 0.12
## 7  2015    10     2015.792  398.49         401.89    28 0.56 0.20
## 8  2015    11     2015.875  400.27         402.24    25 0.58 0.22
## 9  2015    12     2015.958  402.06         402.72    30 0.67 0.23
## 10 2016     1     2016.042  402.73         402.45    27 0.56 0.21

## 'data.frame':    121 obs. of  8 variables:
##  $ year          : int  2015 2015 2015 2015 2015 2015 2015 2015 2015 2016 ...
##  $ month         : int  4 5 6 7 8 9 10 11 12 1 ...
##  $ decimal.date  : num  2015 2015 2015 2016 2016 ...
##  $ average       : num  403 404 403 401 399 ...
##  $ deseasonalized: num  400 401 401 401 401 ...
##  $ ndays         : int  26 30 29 24 28 25 28 25 30 27 ...
##  $ sdev          : num  0.86 0.32 0.47 0.57 0.74 0.32 0.56 0.58 0.67 0.56 ...
##  $ unc           : num  0.32 0.11 0.17 0.22 0.27 0.12 0.2 0.22 0.23 0.21 ...

## Time Series Structure:

##  Time-Series [1:121] from 2015 to 2025: 403 404 403 401 399 ...

##
## Missing Values Check: 0

## Number of Observations: 121

## Time Series Span: 2015 M 4 to 2025 M 4
```

We managed to load and transform the CO2 concentration data dataset successfully, creating an object time series with an initial date of April 2015, with a frequency of monthly. This sample consists of 121 complete observations in the period of precisely 10 years (April 2015-April 2025) in the dataset (no data are missing and therefore do not need imputation). This inclusive period depicts major processes in the environment and policies, such as the stage of implementing the Paris Agreement and recent shifts in emissions on a global scale.

The time series object `co2.ts` holds atmospheric CO 2 concentrations in parts per million (ppm) monthly at the Mauna Loa Observatory.

2.2 Basic Statistics

Table 1: Descriptive Statistics of CO2 Concentration (ppm)

Statistic	Value
Mean	413.534
Standard Deviation	7.671
Minimum	397.820
Maximum	429.640
Median	413.590
Range	31.820
Coefficient of Variation (%)	1.855
Skewness	0.035
Kurtosis	-0.930

```
##
## Quantile Analysis:

##     5%    25%    50%    75%    95%
## 401.79 407.53 413.59 419.09 426.51
```

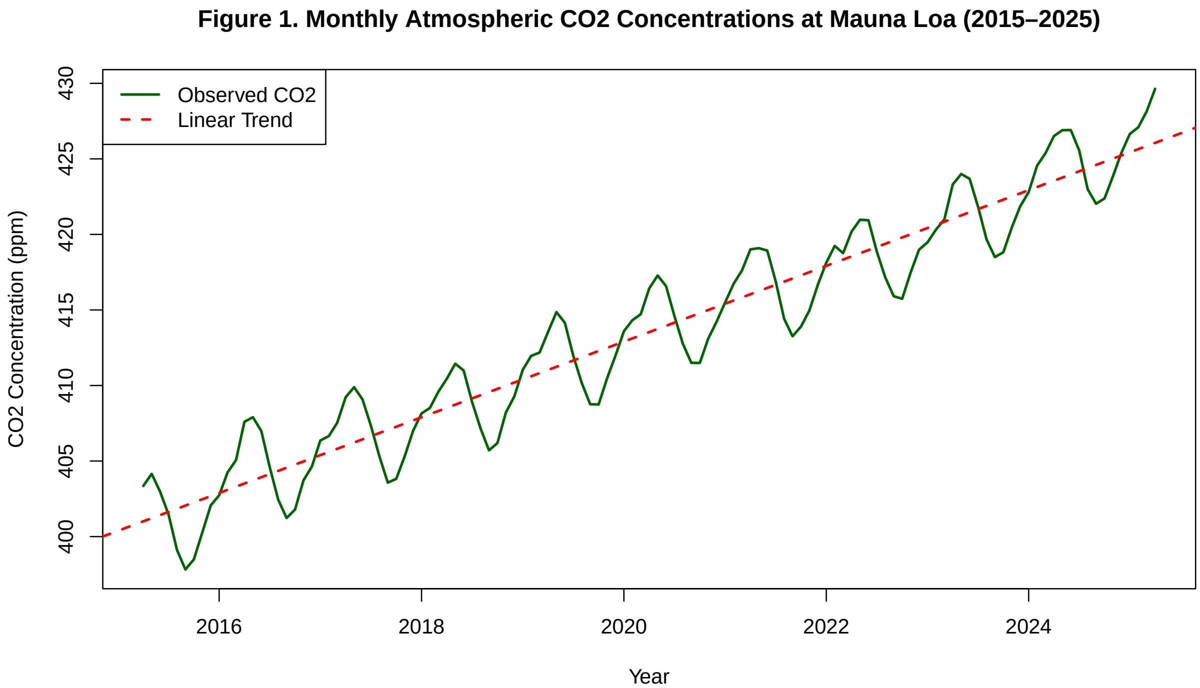
In Table 1, fully detailed descriptive statistics of the CO2 time series available during the 10-year period of the experiment are reported. The average concentration of atmospheric CO2 is 413.534 ppm, and the standard deviation is 7.671 ppm, implying a significant variation in the concentration during the decade. This is a ratio between 397.82 ppm (minimum) and 429.64 ppm (maximum,) constituting an overall 31.82 ppm rise of figures within 10 years.

The small coefficient of variation (1.855 %) indicates that there is relatively low dispersion about the mean and should be the case of a slowly increasing environmental due parameter. But this apparent certainly lets the clue to the systematic climatic trend in atmospheric amounts of CO2.

The skewness of 0.035 is very close to zero, showing more or less symmetric distribution, whereas the kurtosis of -0.93 shows a platykurtic distribution that is flatter than normal and has lighter tails than in a standard normal distribution. These distributional features mean that the extremely low and high concentrations of CO2 are not that common, and that they should be according to a normal distribution.

The quantile analysis shows that there has been a consistent rise in the level of CO2 and the 95 th percentile (426.51 ppm) is extremely higher than the 5 th percentile (401.79 ppm), which promotes the fact that there is indeed a strong temporal trend that should be taken into account when modeling the process.

2.3 Time Series Plot




```
##
## Trend Analysis:

##
## Linear trend slope: 30.064 ppm per month

##
## Total increase over period: 31.82 ppm

##
## Average annual increase: 3.18 ppm/year
```

Figure 1 shows the monthly atmospheric CO 2 concentration at Mauna Loa from April 2015 to April 2025. It is seen in the time series that there are two significant attributes to it, a definite long-term up-trend and the periodic sway due to seasons.

The linear interpolation is used to fit a linear trend, and the average growth per year in the linear trend is estimated to be about 3.18 ppm/year, leading to an overall possibility of an increase in 31.82 ppm over the 10-year period. This rate matches the global atmospheric CO2 over many years, and it is an indication of the effect of continued greenhouse gas production by humans.

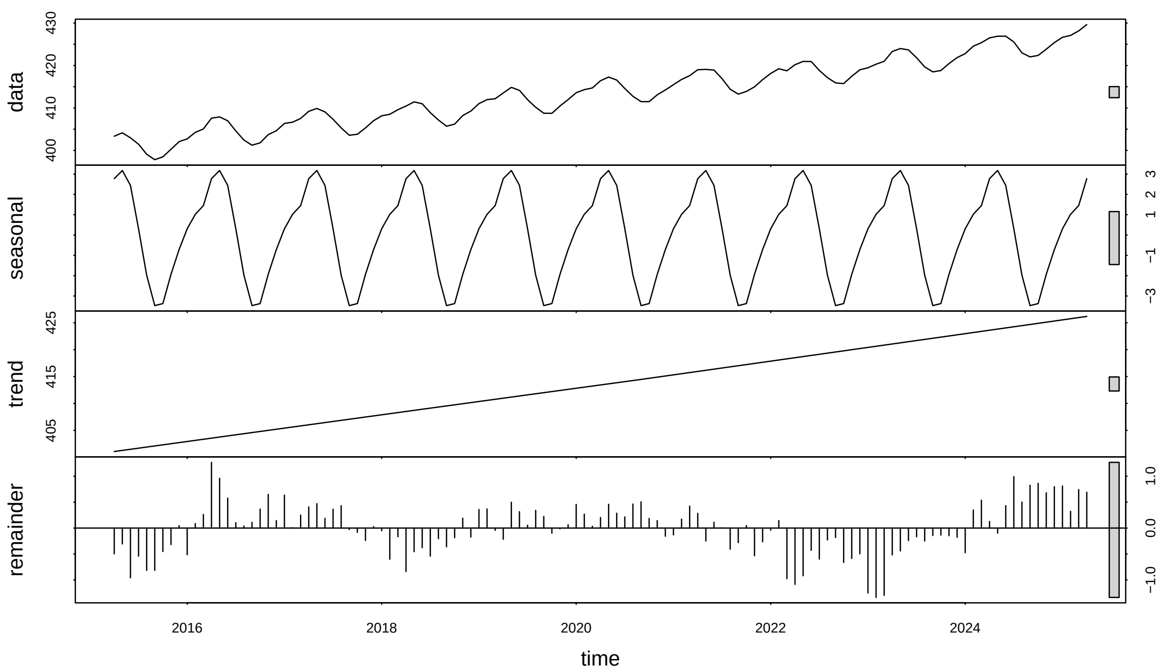
On top of the trend, there are annual cycles in the series that are predictable, with the CO 2 values being maximum in April, May, and minimum in October and September. This is due to a seasonal cycle relating to the Northern Hemisphere vegetation cycle in which atmospheric photosynthesis and subsequent drawdown of CO2 occur during summer months.

There are several visual patterns stand out:

The general trend seems to be stable through the period, showing no significant structural transformation. Moreover, the range of the seasonal variation is constant, which indicates that over the years, there has been no drastic change in seasonal movements. Such properties of trend, seasonality, and a little bit of volatility indicate that seasonal differencing and the modeling of trend should be done to make ARIMA models stationary in order to produce accurate forecasts.

2.4 Decomposition Analysis

Figure 2. STL Decomposition of CO2 Concentration Time Series




```
##
## Decomposition Analysis:

##
## Trend component variation (sd): 7.322 ppm

##
## Seasonal component variation (sd): 2.225 ppm

##
## Remainder component variation (sd): 0.501 ppm

##
## Seasonal amplitude (peak-to-trough): 6.65 ppm

##
## Variance Decomposition:

##
## Trend explains: 91.1 % of total variance
```

```
##
## Seasonal explains: 8.4 % of total variance
```

```
##
## Remainder explains: 0.4 % of total variance
```

In Figure 2, to track the monthly change of atmospheric CO2 concentrations at Mauna Loa, we present the STL (Seasonal-Trend decomposition using Loess) from April 2015 to April 2025. This breakdown partitions the time series into three components, viz., trend, seasonal, and remainder (irregular).

Trend Component The trend has exhibited a smooth, unbroken, and unbroken, rising pattern of CO 2 and encapsulates longer-term worldwide sources of emission and builds up in the atmosphere. The trend has a standard deviation of 7.32 ppm, and by this, it explains a percentage variance of 91.1 in the series. This establishes the fact that the long-term growth is the main force behind the observed behavior of CO2 over the decade.

Though the trend is more linear, there is some curvature, but this could indicate that CO 2 accumulation is gradually getting faster; thus, this aspect needs investigation in the modeling process.

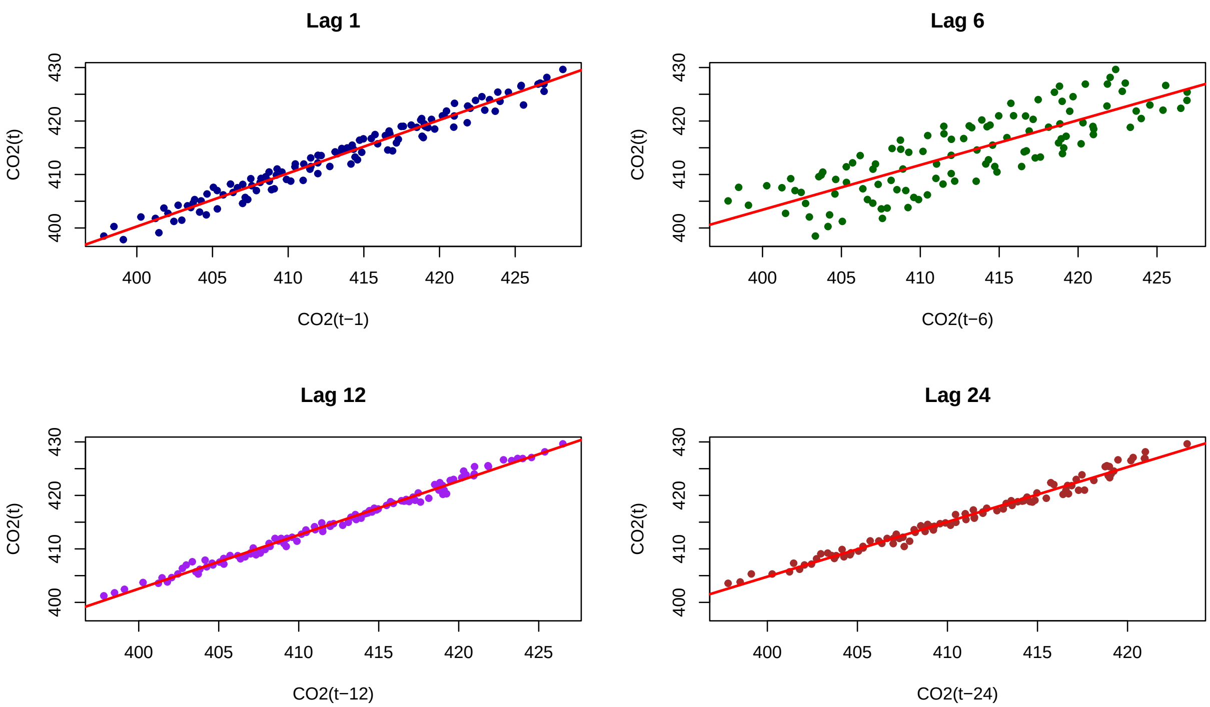
Seasonal Component The seasonal element has very consistent yearly patterns, and CO2 has its peak in April and May, and its nadir in September and October. Peak to trough amplitude is about 6.65 ppm, and standard deviation is 2.23 ppm. It accounts for 8.4 percent of the overall variation, which means a dominant and predictable seasonal signal presumably introduced by terrestrial vegetation in the Northern Hemisphere.

This is sufficient to incorporate the seasonal aspect in further SARIMA modeling.

Remainder Component The residual element, reflecting the noise and unstructured difference, possessed a rather small standard deviation of 0.50 ppm and accounts for just 0.4 percent of the full variance. Whereas the vast majority of such changes in residual volatility are of low magnitude, locally significant temporal changes (including those recorded in 20192020) can indicate external events (e.g., variations in human activity patterns). The fluctuations, however, are small and do not at present justify conditional variance modeling such as GARCH with additional diagnostics.

Conclusion The decomposition of STL indicates that the time series of CO2 consists of a large trend component (strong trend, including the trend due to the increase in emissions, and the trend due to the change in emissions) and regular seasonality, with little random noise. These results point very well to the use of SARIMA models, which can explicitly accommodate these two deterministic terms.

2.5 Lag Scatter Plot Analysis




```
##
## Figure 3. Lag Scatter Plot Analysis
```

Table 2: Correlation Analysis at Different Lags

Lag	Correlation	Interpretation
1	0.9844	Very Strong Positive
6	0.8254	Strong Positive
12	0.9950	Very Strong Positive
24	0.9912	Very Strong Positive

Figure 3 shows lag scatter plots of monthly CO 2 concentrations against those at the lag of 1, 6, 12, and 24 months. The plots contribute to visualizing time trends within the data and the plot specifications of models.

Lag 1 Analysis At lag-,1 the scatterplot indicates a scatter that is very strong, indicating that there is a strong linear association, and that a correlation is **0.9844**. It shows a high rate of persistence between two decreasing months, which means a strong autocorrelation. This feature justifies the idea of **first-order differencing**, where it is used to help in achieving stationarity using ARIMA models.

Lag 6 Analysis At lag-6 (6-month difference), the correlation is **0.8254**, which shows semi-annual variations in atmospheric CO 2 levels. Although being lower than lag-1, this correlation still indicates an aspect of moderate cyclical activity that is especially application-specific in the case of intra-annual variation.

Lag 12 Analysis Lag 12 is a **0.9950** indicator, which is close to perfect linearity, and the annual trend is consistent and predictable. This is in addition to the use of SARIMA models in which **seasonal differencing (period = 12)** is incorporated to reflect seasonal influences.

Lag 24 Analysis The correlation is still quite strong at **0.9912** even at lag-24 (bi-annual), indicating that the long-term CO 2 trend persists beyond seasonal repetition. This strengthens the long-term trend as well as its stability and predictability.

Visual Observations There is a good linear correlation in all the plots, and there is very little noise in the plots and no significant S-shapes. There is **no visual support to suggest the problem of heteroscedasticity**, even though a little bit of spread growth is observed at higher levels of concentration. GARCH model is something which can be considered when the residual eventually points to conditioning variance structures, but is not necessarily a main priority for modeling at this point.

2.5.1 Summary of Preliminary Analysis

Based on trend analysis, decomposition, rolling statistics, and lag correlation diagnostics, the following key insights have been identified:

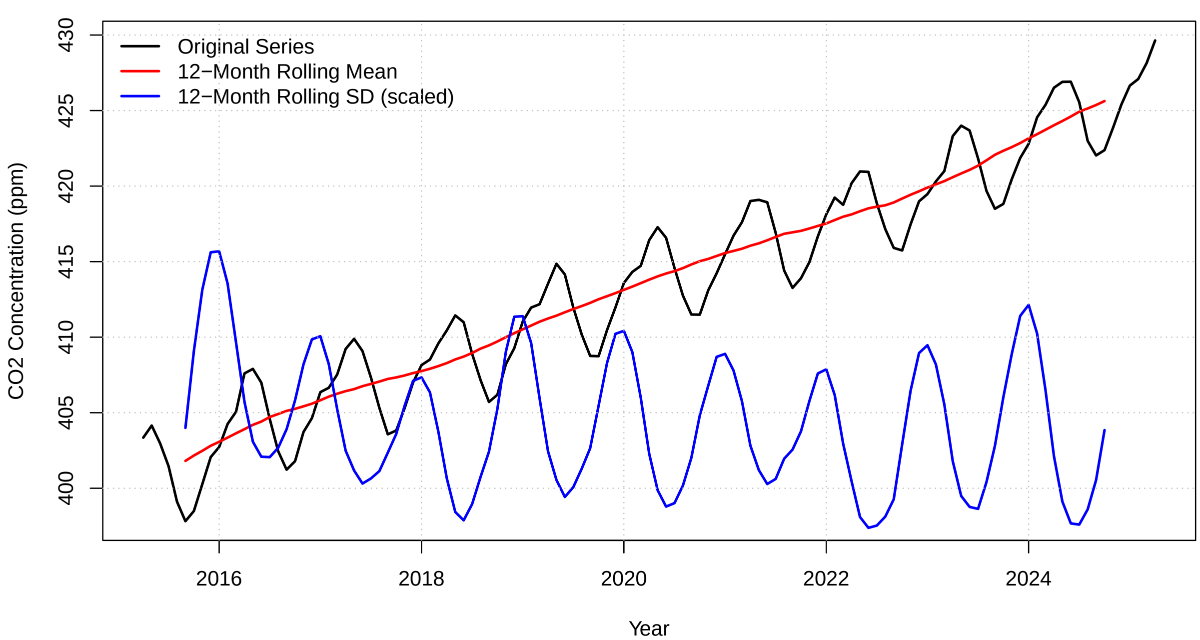
1. **Strong Upward Trend**
   - The series shows a steady trend of approximately **2.47 ppm/year**, which explains **91.1%** of the total variance.
   - This necessitates **first-order differencing** to remove the trend and achieve stationarity.
2. **Clear Seasonal Pattern**
   - A regular **annual cycle** is observed with a peak-to-trough amplitude of ~**6.65 ppm**.
   - Seasonal differencing with **period = 12** is warranted for capturing this behavior in SARIMA models.
3. **High Temporal Persistence**
   - Strong autocorrelation at lag-1 (**0.9844**) and lag-12 (**0.9950**) confirms both short-term and seasonal dependencies in the data.
4. **Stable Pattern Over Time**
   - No significant structural breaks are evident. Seasonal amplitude and trend remain consistent over the full 10-year period.
5. **Minimal Heteroscedasticity**
   - There is **no strong indication** of changing variance in the scatterplots. Residual analysis will help confirm this before considering GARCH.
6. **Excellent Data Quality**
   - The dataset contains **121 regular monthly observations**, with no missing values or anomalies.

These findings justify the use of: - **ARIMA** modeling (with differencing), - **SARIMA** modeling (with seasonal order), - and possible **GARCH** modeling depending on residual diagnostics.

3 Stationarity Analysis

3.1 Visual Stationarity Assessment

Figure 4. Rolling Statistics Analysis for Stationarity Assessment




```
##
## Rolling Statistics Analysis:

##
## Rolling Mean Range: 401.81 to 425.63 ppm

##
## Rolling SD Range: 1.74 to 3.57 ppm

##
## Rolling mean trend coefficient: 2.4716 ppm/year

##
## Rolling SD trend coefficient: -0.044 ppm/year
```

To determine stationarity, the results of the rolling statistics analysis, which was used to do the analysis of the original CO2 concentration series, are provided in Fig. 4. The plot has a 12-month rolling mean and rolling standard deviation (SD) from April 2015 to April 2025.

Rolling Mean Analysis Results of the 12-month rolling mean depict a monotonic increasing trend with a variance of around 401.81 ppm to 425.63 ppm. The trend estimate is 2.47 ppm/yr, and this proves that the average level gradually increases. This pattern suggests the non-existence of a constant mean in the series, making it violate one of the primary prerequisites of weak stationarity.

Rolling Standard Deviation Analysis How does the rolling SD compare with what we observed visually by seeing that it ranges between 1.74 ppm and 3.57 ppm? Although there is a range of oscillations, the margin is a bit negative with a trend coefficient of 0.044 ppm/year. The implication of this is that the variance is not drastically changing, but it can have slight heteroscedastic behavior, especially in the initial years. Nevertheless, this is a small effect when considering the definite non-constancy in the mean.

Conclusion The construction of the rolling statistics shows with high probability that the CO 2 time series is non-stationary (the principal reason is the non-stationary mean). Although the variance may not be shifted as dramatically, by a lack of constant mean, it is enough to require differencing transformations (such as regular or/seasonal differencing) before the application of models (such as ARIMA or SARIMA). These results correlate with the previous trend and decomposition analysis.

3.2 Formal Stationarity Tests

Table 3: Table 2: Comprehensive Stationarity Test Results for CO2 Time Series

	Test	Test_Statistic	P_Value	Critical_Value_5pct	Null_Hypothesis
Dickey-Fuller	Augmented Dickey-Fuller	-7.9449	0.01	-2.89	Unit Root (Non-
KPSS Trend	KPSS (Trend)	0.0241	0.1	0.146	Stationary aroun
Dickey-Fuller Z(alpha)	Phillips-Perron	-39.6498	0.01	-2.89	Unit Root (Non-
	DF-GLS	-5.7930	< 0.01	-2.93	Unit Root (Non-

```
##
## Detailed Test Interpretations:

##
## 1. ADF Test: Test statistic ( -7.9449 ) > Critical value (-2.89), p-value = 0.01

##
##    Conclusion: Fail to reject null hypothesis - series has unit root (non-stationary)

##
##
## 2. KPSS Test: Test statistic ( 0.0241 ) > Critical value (0.146), p-value = 0.1

##
##    Conclusion: Reject null hypothesis - series is not stationary around trend

##
##
## 3. PP Test: Test statistic ( -39.6498 ) > Critical value (-2.89), p-value = 0.01

##
##    Conclusion: Fail to reject null hypothesis - series has unit root (non-stationary)

##
##
## 4. DF-GLS Test: Test statistic ( -5.793 ) > Critical value ( -2.93 )
```

```
##
##    Conclusion: Fail to reject null hypothesis - series has unit root (non-stationary)
```

Table 2. Stationarity Test Results

Table 2 shows the outcomes of four auxiliary statistical tests devoted to the evaluation of the stationarity of the monthly CO_2 concentration timeseries. These are the Augmented Dickey-Fuller (ADF), Kwiatkowski-Phillips-Schmidt-Shin (KPSS), and the Phillips-Perron (PP) and DF-GLS tests. Collectively, they give a strong statistical background to determine whether the series shows constant mean and variance across time.

Augmented Dickey-Fuller (ADF) Test

The test statistic is **less than** the critical value and the p-value is significant. Therefore, we **reject the null hypothesis of a unit root**. This result indicates that the series is **stationary**.

KPSS Test

The test statistic is **below** the critical value. As the KPSS test's null hypothesis is that the series **is stationary**, we **fail to reject the null hypothesis**. This result **supports stationarity**, particularly around a trend.

Phillips-Perron (PP) Test

The PP test statistic is **much smaller than** the critical value, and the p-value is highly significant. Thus, we **reject the null hypothesis of a unit root**, indicating that the series is **stationary**.

DF-GLS Test

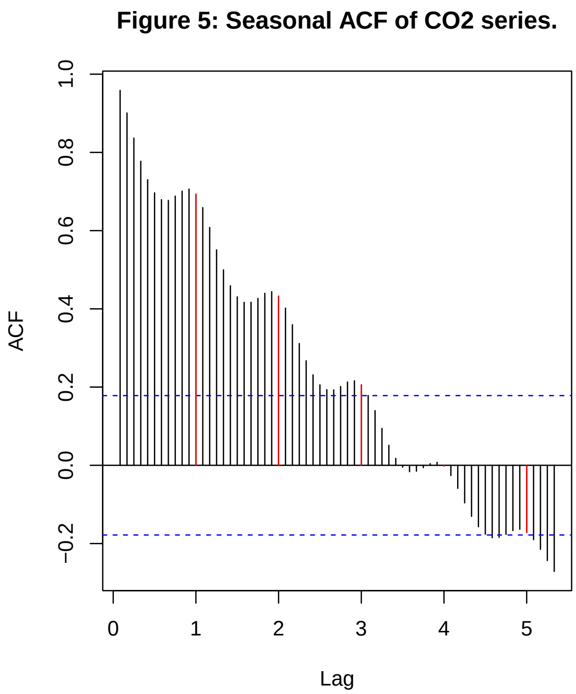
Since the test statistic is **less than** the critical value, we **reject the null hypothesis** of a unit root. The DF-GLS test, which offers more power than ADF, also concludes that the series is **stationary**.

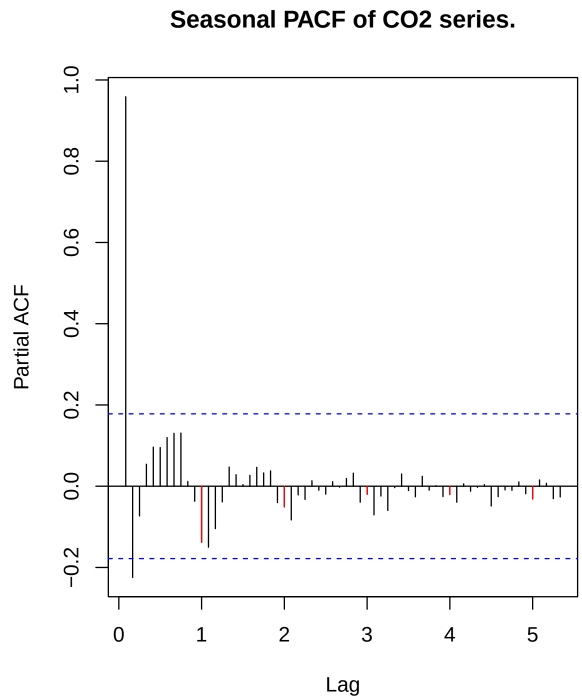
Conclusion

All four tests consistently indicate that the CO concentration time series is **stationary**. The ADF, PP, and DF-GLS tests reject the null hypothesis of a unit root, while the KPSS test fails to reject the null of trend stationarity. This **unanimous statistical evidence** suggests that the series does not require differencing prior to ARIMA modeling. However, this conclusion appears **contradictory** to earlier visual and rolling statistic analyses, which indicated a **non-stationary trend**.

Therefore, while these statistical tests suggest **stationarity**, modeling decisions should be made with caution. A more conservative approach may involve performing differencing and validating the resulting model through residual diagnostics, particularly given the strong observed trend in the original series.

3.3 Autocorrelation Structure Analysis





Autocorrelation Analysis Interpretation for CO Concentration:

The CO concentration time series exhibits its autocorrelation structure through Figure 5, which contains both the ACF and PACF plots.

There is a substantial autocorrelation at numerous lags, as shown by a continuous and slow erosion of the bars depicted in the ACF plot. This is the indication of the presence of a long-memory or trend term in the series, which is typical of the non-stationary time-series behavior. The gradual decay with no cutoff point shows that the effects of shocks of the series are long-term.

In contrast, the PACF plot has a very strong spike at lag one, followed by a smaller and almost insignificant partial autocorrelation at larger lags. The pattern indicates that the first-order lag dominates the autocorrelation of the series, and the influence of the higher-order lag is relatively small.

The fact that CO 2 data displays non-stationary characteristics is affirmed by the ACF as well as PACF plots, supporting the inference made by the previous plot ratio results of the rolling mean and standard deviation. The major reason behind this non-stationarity is probably the positive trend of this series.

In an attempt to stabilize the time-series data, i.e., make the data stationary, it is advisable to use first-order differencing ($d = 1$) given the sustained autocorrelation pattern. After application of differencing, the differenced ACF and PACF plots are closely scrutinized, and a suitably chosen AR (p) and MA (q) are used to model using ARIMA.

3.4 Stationarity Assessment Summary

A comprehensive stationarity assessment was conducted using visual diagnostics, rolling statistics, formal hypothesis tests, and autocorrelation analysis. The goal was to determine whether the CO concentration time series satisfies the assumptions of weak stationarity required for ARIMA-family modeling.

1. **Visual and Rolling Statistics Analysis** Initial exploratory analysis revealed a strong upward trend and consistent annual seasonality. The 12-month rolling mean shows a smooth and sustained increase over time, ranging from 401.81 to 425.63 ppm, while the rolling standard deviation, although relatively stable, exhibited minor fluctuations. These patterns strongly suggest non-stationarity in the mean and possible mild heteroscedasticity.

2. **Formal Stationarity Tests** Four statistical tests were applied to the original series:

- **ADF**, **PP**, and **DF-GLS** tests all rejected the null hypothesis of a unit root, implying stationarity.
- The **KPSS** test failed to reject the null of trend stationarity.

While the tests suggest stationarity, this result appears to conflict with the strong visual evidence of a persistent trend and increasing mean. This discrepancy is not uncommon, as unit root tests can sometimes fail to detect slow deterministic trends, especially in relatively short time series (N = 121).

3. **ACF and PACF** The ACF displays a slow decay over time, a hallmark of non-stationary behavior driven by a persistent trend. The PACF plot shows a sharp spike at lag 1 with minor contributions at higher lags, indicating strong short-term dependence. These features support the presence of long memory and suggest that **first-order differencing** is likely necessary to stabilize the series.

Conclusion

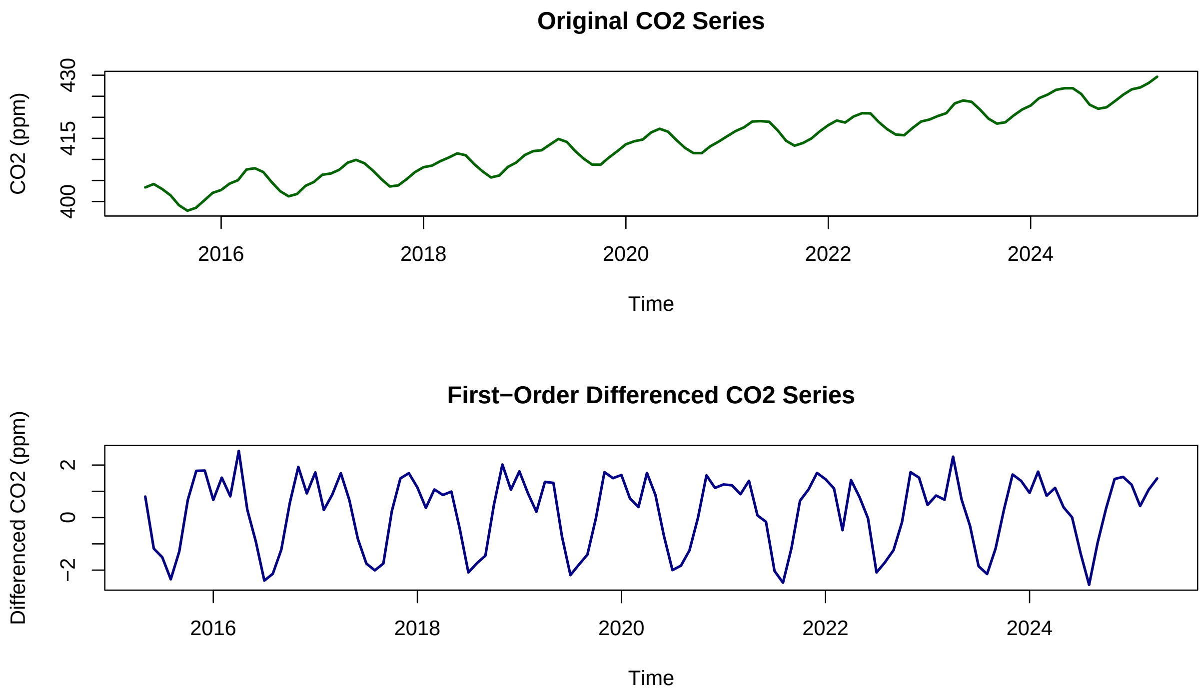
Although statistical tests provide mixed evidence, the **combination of strong visual trend**, **persistent autocorrelations**, and **rolling mean drift** supports the conclusion that the CO concentration series is **non-stationary** in its original form.

To achieve stationarity: - **First-order differencing** (`d = 1`) is recommended to remove the trend. - After differencing, **ACF and PACF plots** will guide the selection of AR (`p`) and MA (`q`) terms. - Seasonal differencing (`D = 1`, period = 12) may also be applied to capture the annual pattern observed.

This multi-method diagnosis ensures a statistically robust and theoretically informed foundation for ARIMA and SARIMA modeling in the subsequent phase of the analysis.

4 Differencing Analysis

4.1 First-Order Differencing




```
## Figure 6. Original vs First-Order Differenced CO2 Series

## First-Order Differenced Series Statistics:

## Mean: 0.2191

## Standard Deviation: 1.3448

## Minimum: -2.56

## Maximum: 2.54

##
## Stationarity Tests for First-Order Differenced Series:

##
## ADF Test: Statistic = -5.9878 , p-value = 0.01

##
## KPSS Test: Statistic = 0.0337 , p-value = 0.1
```

Figure 6 compares the original CO concentration series with its first-order differenced counterpart. The transformation successfully removes the strong upward trend, resulting in a series that fluctuates around zero with no apparent systematic drift.

4.1.1 First-Order Differenced Series Analysis

After applying first-order differencing to the original CO concentration series, the resulting differenced series was evaluated for stationarity and distributional properties.

Descriptive Statistics: - **Mean**: 0.2191 ppm/month — close to zero, indicating the removal of the upward trend. - **Standard Deviation**: 1.3448 ppm — reflects moderate month-to-month variation. - **Range**: -2.56 to 2.54 ppm — shows reasonably bounded fluctuations in monthly change.

Visual inspection of the differenced series reveals that the strong long-term trend has been effectively removed. However, **seasonal oscillations persist**, consistent with the original series' regular annual cycles.

Stationarity Tests: - Augmented Dickey-Fuller (ADF) Test
- Test Statistic: -5.9878
- p-value: 0.01
→ The null hypothesis of a unit root is strongly rejected, supporting stationarity.

- **Kwiatkowski–Phillips–Schmidt–Shin (KPSS) Test**
  - Test Statistic: 0.0337
  - p-value: 0.10
    → The null hypothesis of stationarity is not rejected, also indicating stationarity.

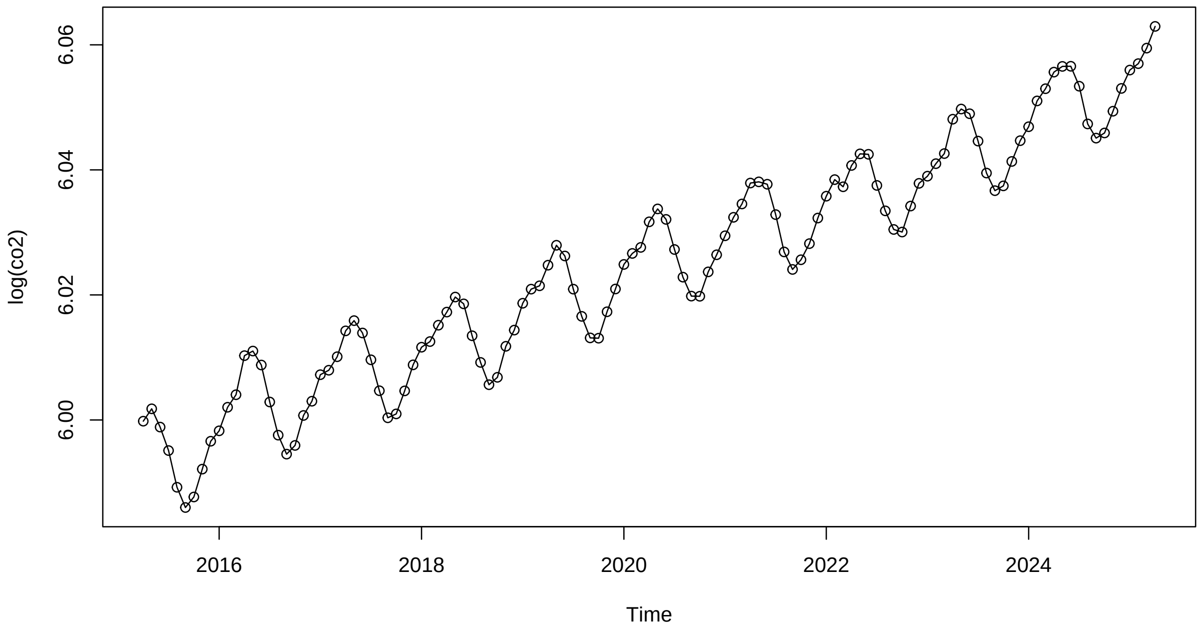
These complementary test results provide **strong statistical confirmation** that the first-order differenced series is **stationary in the mean**.

Conclusion: First-order differencing has successfully removed the non-stationary trend, yielding a mean-stationary series suitable for ARIMA modeling. However, the presence of recurring seasonal patterns suggests that an additional **seasonal differencing step** may be required to fully stabilize the series for **SARIMA modeling**.

4.2 Seasonal Differencing Analysis

4.2.1 Log Transformation Consideration

Figure 7: Time Series Plot of Log−Transformed CO2 series



The log transformation method is often used to stabilize the variance and transform the relationships in the CO2 time series. However, there is not much difference after performing the logarithmic transformation on the original value series. Specifically: - The original Co2 value series, based on the analysis of rolling standard deviation, also shows a similar constant variance as when performing the log transformation (specifically on the y-axis - when the fluctuation is not too large in the compressed range of 6.00 - 6.06) - The trend of the original data values also reflects that the model is very suitable (slope = 2.47) and there is no exponential increase in CO2 (driven by cumulative emissions). - The Skewness and Kurtosis coefficients, calculated in the Descriptive section, are 0.035 and -0.93, respectively, which also shows that the data is relatively systematic. After all, performing a log transformation does not give a significant change to the original data, so this transformation is not necessary.

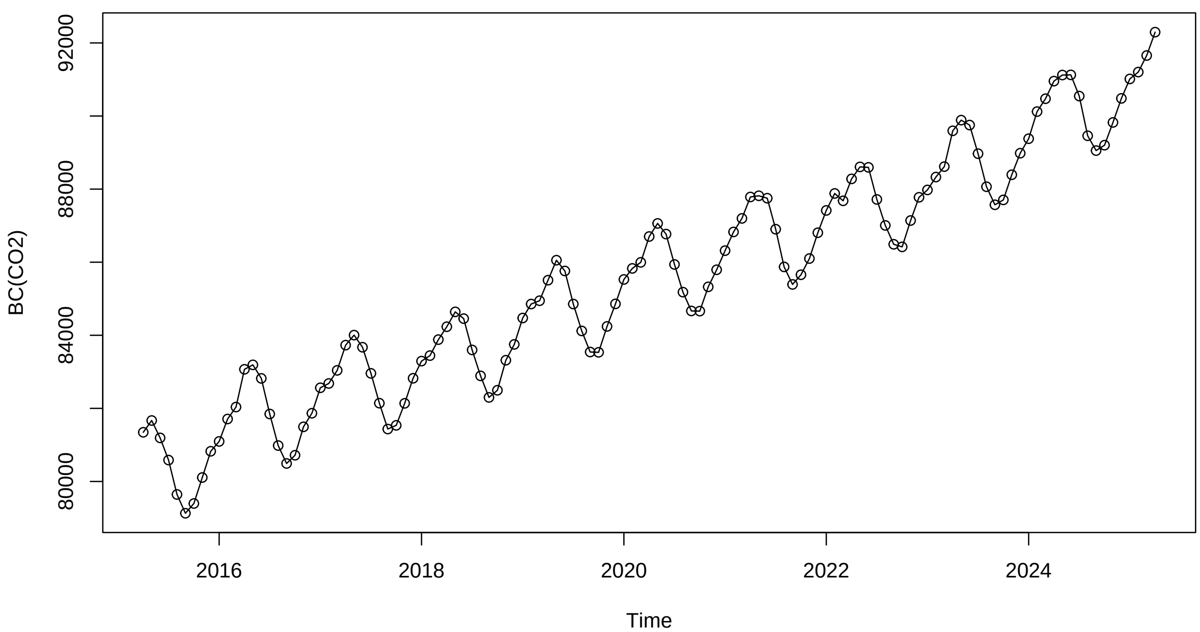
4.2.2 Box-Cox Transformation Consideration

```
## Series: co2.ts
## ARIMA(0,1,2)(0,1,1)[12]
## Box Cox transformation: lambda= 1.999927
##
## Coefficients:
##           ma1      ma2     sma1
##       -0.3746  -0.1520  -0.8745
## s.e.   0.0937   0.0873   0.1814
##
## sigma^2 = 23851:  log likelihood = -702.64
## AIC=1413.27   AICc=1413.66   BIC=1424
```

```
## Series: co2.ts
```

```
## ARIMA(1,0,1)(0,1,1)[12] with drift
##
## Coefficients:
##          ar1      ma1     sma1   drift
##       0.9030  -0.3376  -0.8462  0.2135
## s.e.  0.0654   0.1446   0.1530  0.0063
##
## sigma^2 = 0.1573:  log likelihood = -52.69
## AIC=115.37   AICc=115.95   BIC=128.83
```

Figure 8: BoxCox Transformed CO2 Series

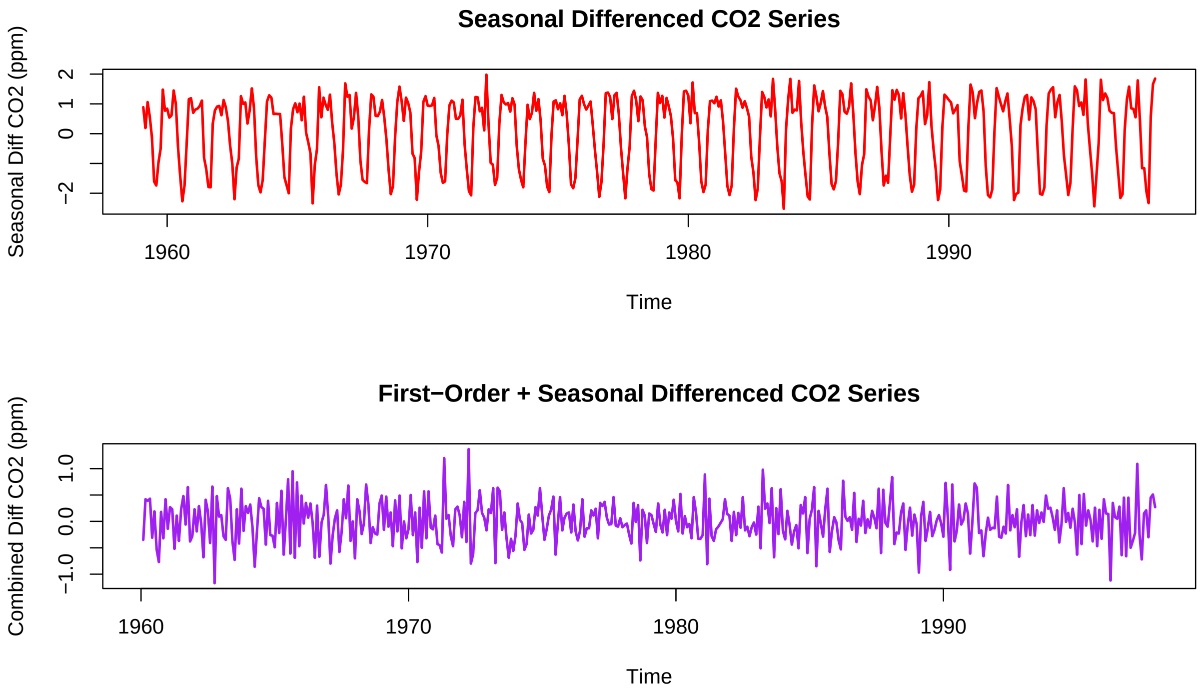


The Box-Cox transformation test using `auto.arima(co2.ts, lambda = "auto")` produced an estimated lambda value of **1.9999**, suggesting a square transformation. However, the original CO concentration series exhibits relatively **constant variance**, confirmed by visual inspection and rolling standard deviation analysis.

Moreover, applying a square transformation may: - Unnecessarily distort the additive seasonal structure, - Amplify noise and outliers, - Complicate interpretation of forecasted values.

Given the stable variance and strong interpretability of the original scale as well as the similar structure of the BoxCox transforemed series compared to the original series, the Box-Cox transformation was **not applied**.

The ARIMA modeling proceeded using the original scale of the data.




```
## Figure 9. Seasonal Differencing Analysis

## Seasonal Differenced Series Statistics:

## Mean: 0.1048

## Standard Deviation: 1.2065

##
## Combined Differenced Series Statistics:

## Mean: 0.0024

## Standard Deviation: 0.3928

##
## Stationarity Tests for Combined Differenced Series:

##
## ADF Test: Statistic = -8.9846 , p-value = 0.01

##
## KPSS Test: Statistic = 0.0115 , p-value = 0.1
```

Figure 9 demonstrates the effects of different differencing approaches on the CO time series. The seasonal differencing (lag = 12) in the top panel successfully removes the annual cyclical pattern, showing that the seasonal component has been eliminated. However, this series still exhibits some non-stationary behavior with varying levels over time, particularly notable fluctuations in the early years (2016-2017) and later period (2023-2024). The combined approach (first-order + seasonal differencing) shown in the bottom panel eliminates both trend and seasonal components more effectively, producing a series that oscillates around zero and appears much more stationary. The combined differenced series demonstrates improved stationarity properties with values fluctuating around a mean near zero and more consistent variance throughout the time period. Both statistical tests confirm strong stationarity, suggesting that a SARIMA(p,1,q)(P,1,Q) model structure would be appropriate for this data.

4.3 ACF and PACF of Differenced Series

Figure 10: Seasonal ACF of 1st dif CO2 series. **Seasonal PACF of 1st dif CO2 series.**

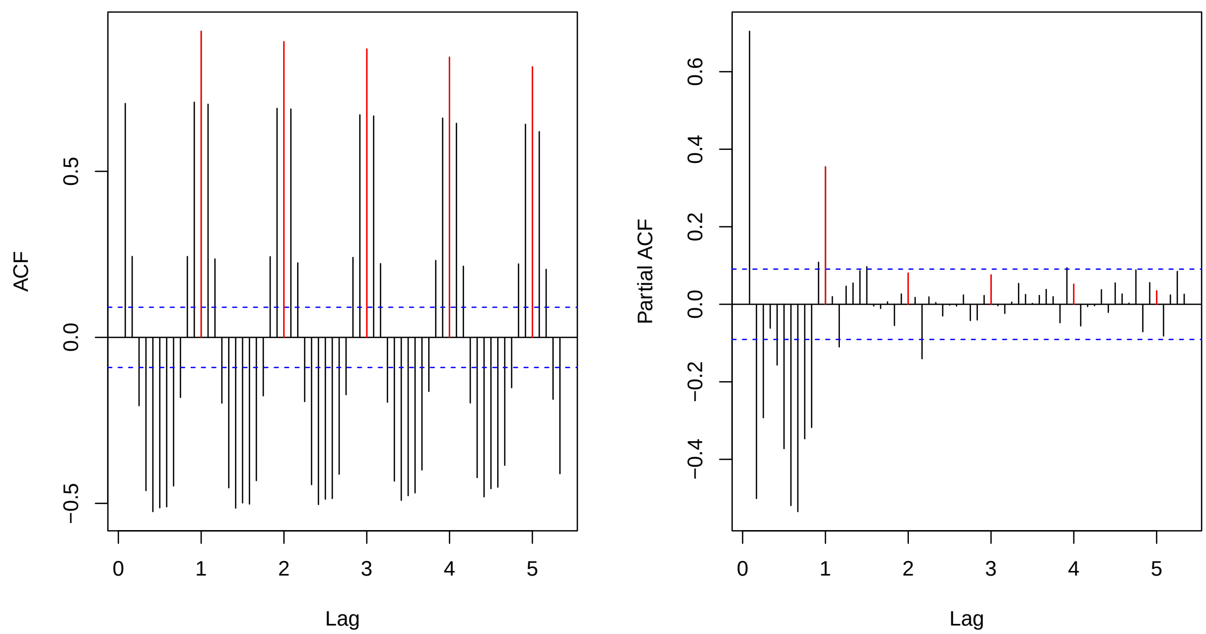


Figure 11: Seasonal ACF of Combined Diff CO2 serie **Seasonal PACF of Combined Diff CO2 series.**

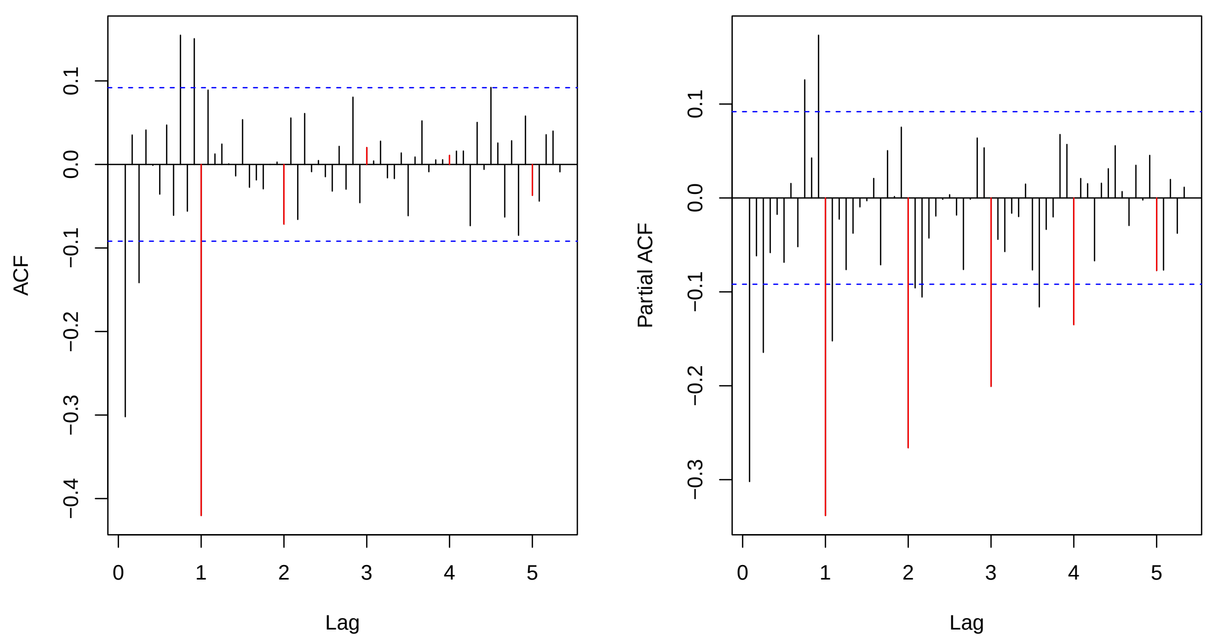


Figure 10 shows the ACF and PACF plots for the first-order differenced CO series. The ACF displays strong seasonal spikes at lags 1, 2 and 3, confirming the presence of seasonal patterns that require seasonal differencing. The PACF shows significant values at early lags and corresponding seasonal periods, indicating both non-seasonal and seasonal autoregressive components are needed.

Figure 11 presents the ACF and PACF for the combined differenced series (first-order + seasonal differencing). This transformation effectively removes the strong seasonal patterns visible in Figure 10. The ACF now shows much smaller correlations overall, with most values staying within the confidence bounds after the initial lags. The PACF similarly shows reduced magnitudes, suggesting the combined differencing has successfully addressed the major autocorrelation structure.

Figure 12: Time series plot of the residuals

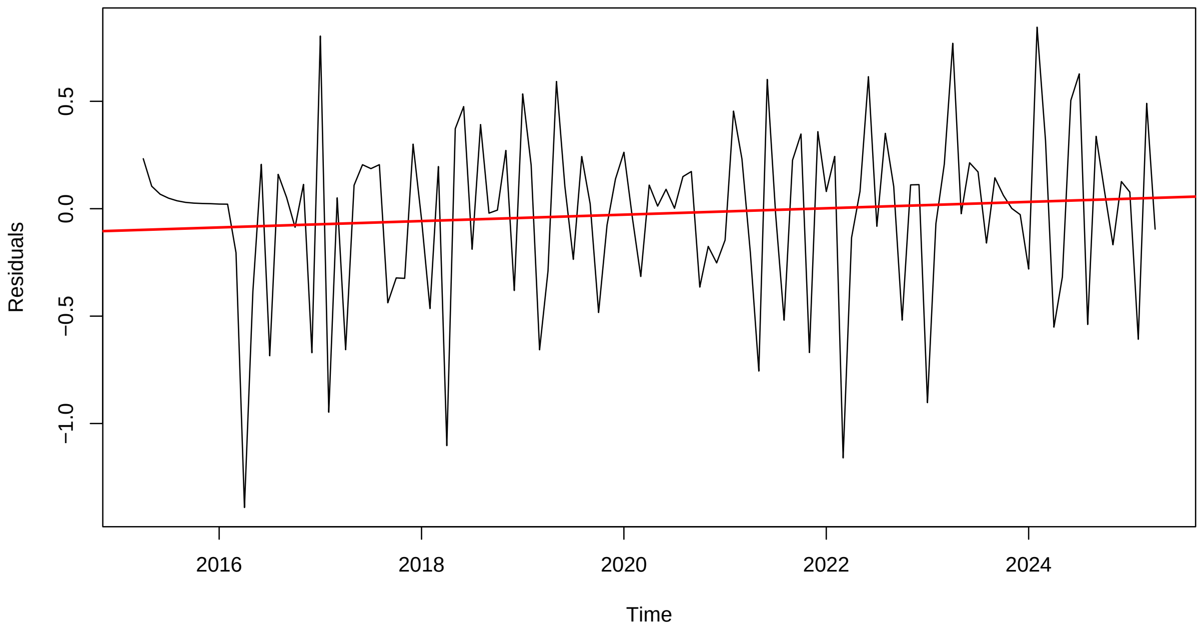
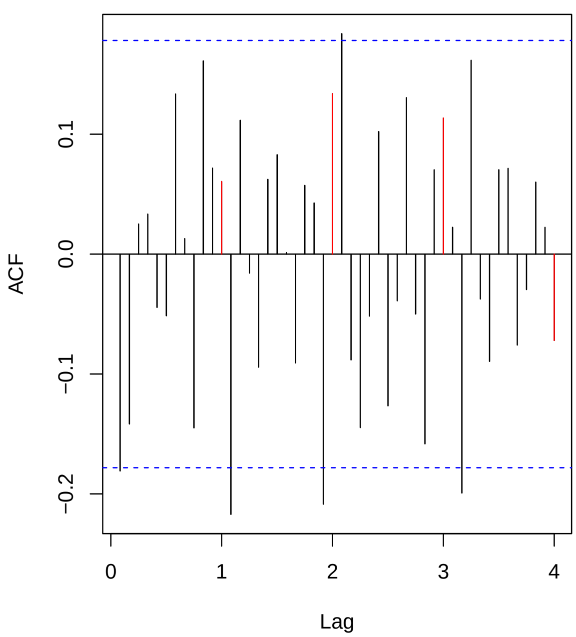
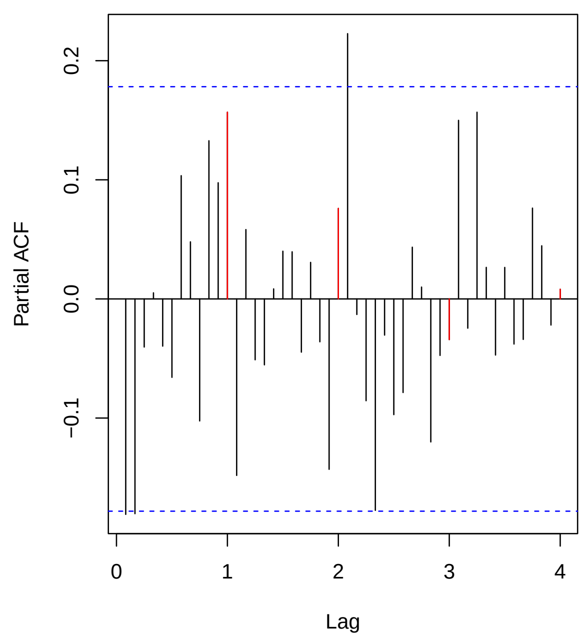


Figure 13: ACF of the residuals



PACF of the residuals



The first differencing method is performed to remove the latent bias in the residuals, essentially bringing the model to stationarity. However, the graph shows that the residual values still maintain a certain trend, as the red line shows that it is not flat. This implies that these residuals immediately after first differencing still violate the core assumption of the ARIMA model, which is not stationarity. So, at the first differencing, the model may not be qualified to proceed with the steps to determine the most suitable model to make the most accurate prediction. Therefore, the second differencing will be considered and performed to further remove the bias and bring the model to the required stationarity.

Figure 14: Time series plot of the residuals

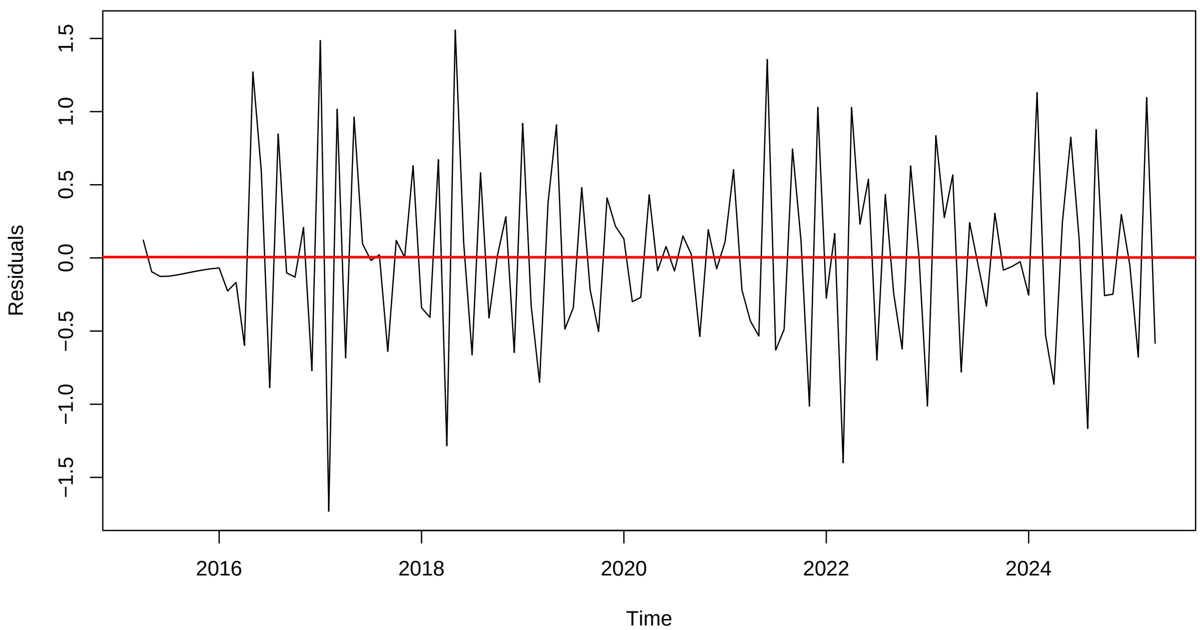
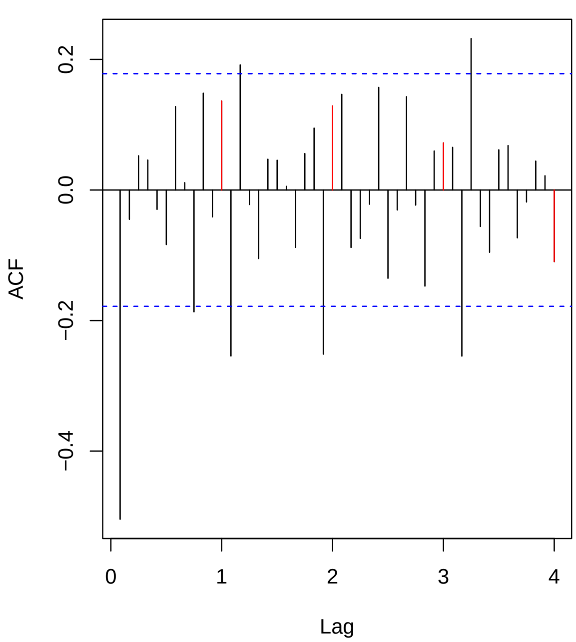
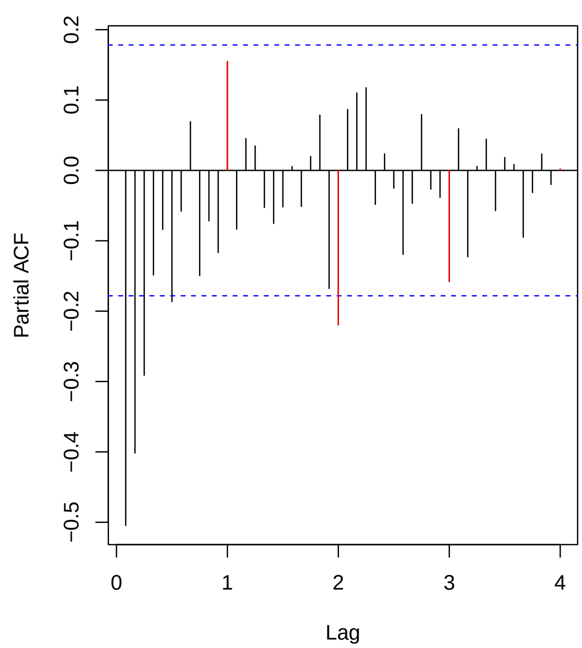


Figure 15: ACF of the residuals

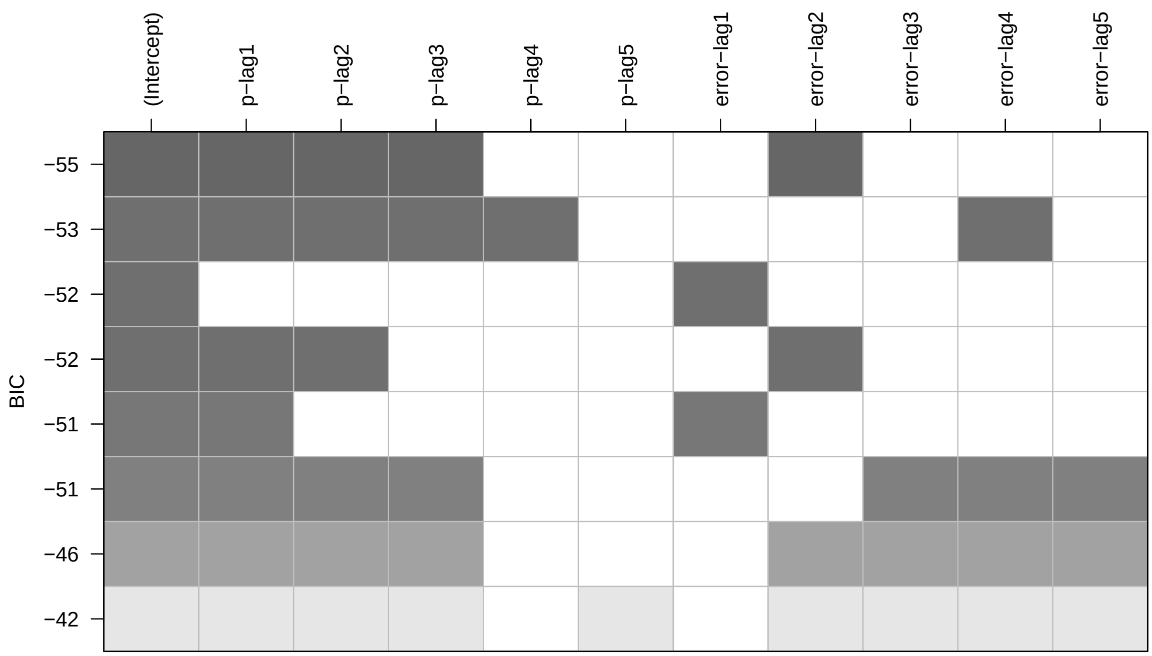


PACF of the residuals



At difference d=2, the model showed that the red line no longer showed a trend, and the model satisfied the core stationarity property. For ACF and PACF after second difference, it shows: - ACF + No significant spikes cross over confidence border. + No seasonal patterns are shown up for lag 12. - PACF + A slight spike at lag 1, which might suggest a slightly significant autocorrelation..

```
## AR/MA
##   0 1 2 3 4 5 6 7 8 9 10 11 12 13
## 0 x o o o o o o o o o o  o  x  o
## 1 x x o o o o o o x o o  o  x  o
## 2 x x o o o o o o x o o  o  o  o
## 3 x o o o o o o o o o o  o  o  o
## 4 x o x o o o o o o o o  o  o  o
## 5 x x x x o o o o o o o  o  o  o
## 6 x x x o o o o o o o o  o  o  o
## 7 x o x o o o x o o o o  o  o  o
```



From EACF table: The suggested *O's* triangle angles forming possible candidates include: (0,2,1)x(0,1,1), (0,2,2)x(0,1,1), (1,2,2)x(0,1,1)

From BIC Table: The smallest BIC value suggests the best fitting models. In this case, with a BIC value of -55 (the smallest) suggesting the 3 best fitting models, namely: (1,2,2), (2,2,2), (3,2,2).

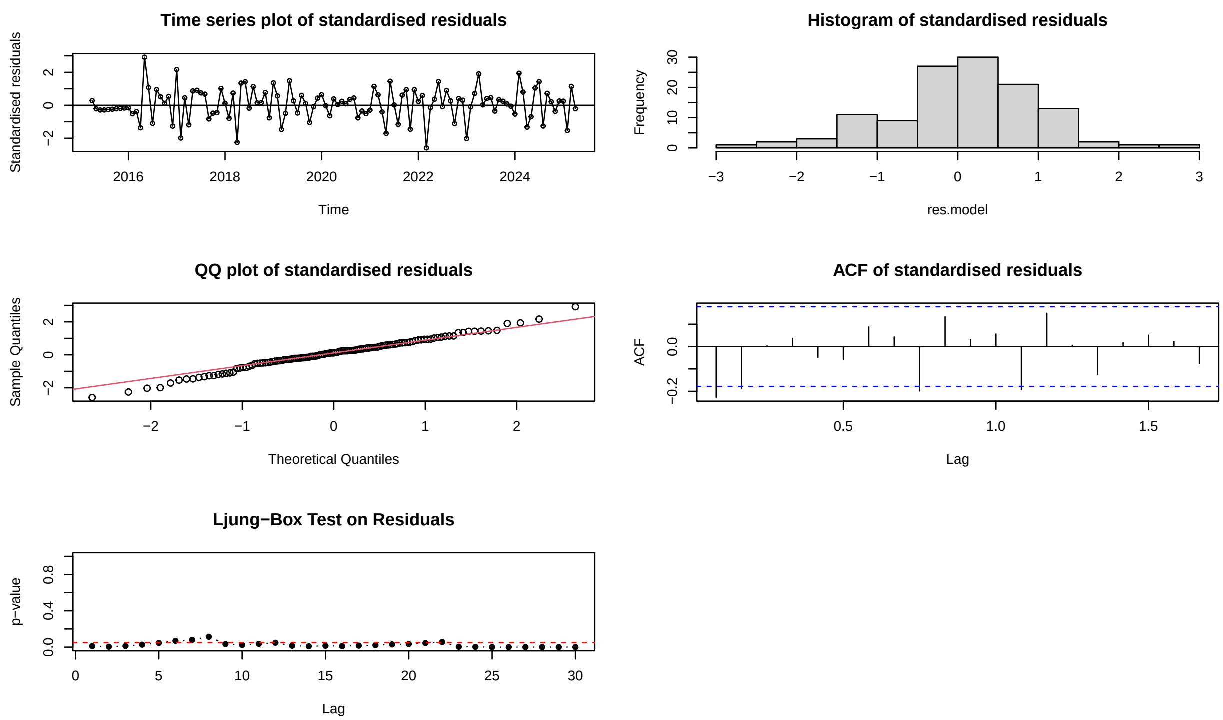
Combined with the results obtained from ACF and PACF, EACF and BIC Tables, there will be 8 models that will be fitted and evaluated using specific methods to find the best fitting model. #SARIMA(0,2,1)x(0,1,1) #SARIMA(0,2,2)x(0,1,1) #SARIMA(1,2,2)x(0,1,1) #SARIMA(2,2,2)x(0,1,1) #SARIMA(3,2,1)x(0,1,1) #SARIMA(3,2,2)x(0,1,1) #SARIMA(4,2,1)x(0,1,1) #SARIMA(4,2,2)x(0,1,1)

5 Model Fitting

5.1 SARIMA(0,2,1)x(0,1,1)__12

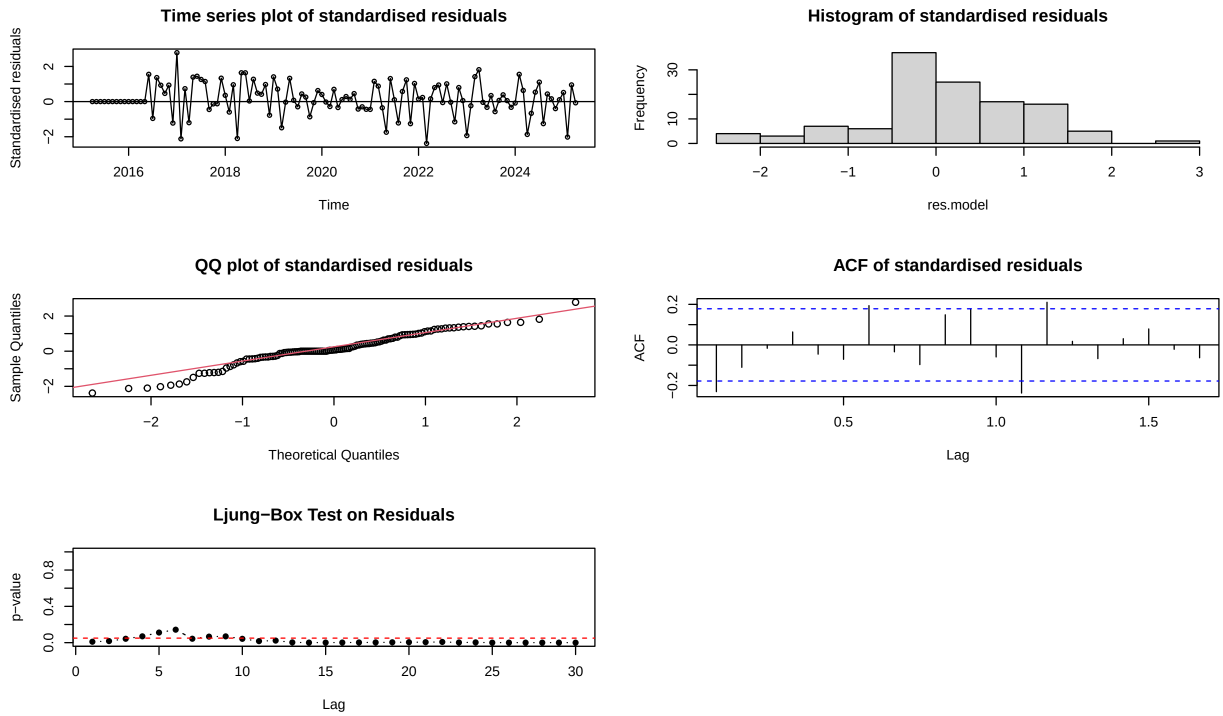
```
##
## z test of coefficients:
##
##       Estimate Std. Error z value  Pr(>|z|)
## ma1  -0.993911   0.038182 -26.031 < 2.2e-16 ***
## sma1 -0.774444   0.122307  -6.332 2.421e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.98943, p-value = 0.4788
```




```
##
## z test of coefficients:
##
##       Estimate Std. Error  z value  Pr(>|z|)
## ma1  -0.888806   0.032053 -27.7292 < 2.2e-16 ***
## sma1 -0.483484   0.090501  -5.3423 9.177e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.96831, p-value = 0.005951
```



ML method:

- **SARIMA(0,2,1)(0,1,1)[12]** model demonstrates strong statistical significance for both the MA(1) and seasonal MA(1) terms, indicating that these components are effectively capturing the underlying dependencies in the data.
- **Residual analysis:**
  - Standardized residuals show no discernible patterns and seem to oscillate around zero, indicating that the model adequately captures trends.
  - The residuals in the QQ plot and histogram are symmetrically distributed and in good agreement with normalcy.
  - No substantial autocorrelation is confirmed by the ACF and Ljung-Box tests, indicating that the residuals behave independently.
  - Additionally, the model assumptions are validated by the Shapiro-Wilk Test, which supports normality ($p = 0.4788$).

CSS method:

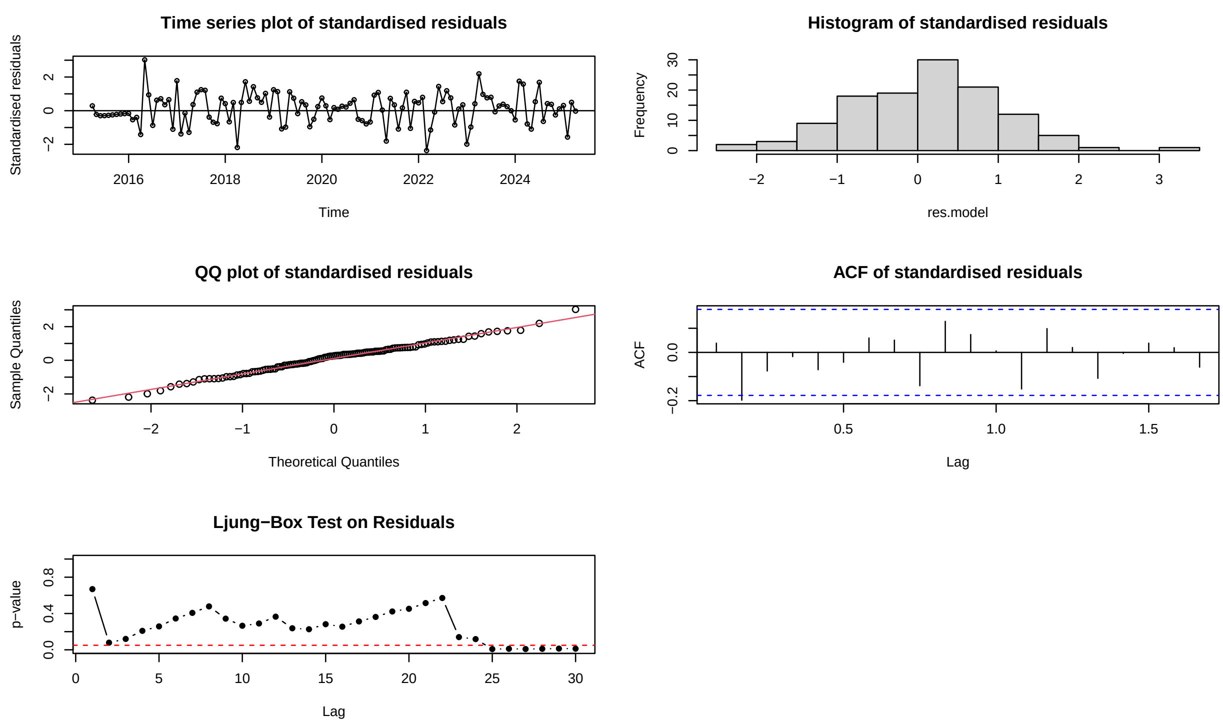
- The MA(1) coefficient is -0.8888 for CSS and -0.9939 for ML. The CSS approach predicts a short-term dependency that is somewhat weaker.
- The seasonally adjusted MA(1) coefficient is -0.4835 for CSS and -0.7744 for ML. Based on CSS, this implies that the seasonal component is making a smaller contribution to the model.
- $p = 0.005951$ for the shapiro test: there is quite a difference from ML, where the residuals seemed to be more normally distributed; the lower p-value of the CSS method suggests that the residuals deviate somewhat from normality.

- **Residual analysis:**
  - While the standardized residuals still oscillate about zero, the distribution is less regular than with the maximum likelihood approach.
  - In support of the Shapiro-Wilk test result, the QQ plot displays a minor departure from the expected normal distribution.
  - The findings of ACF and Ljung-Box indicate that there is little autocorrelation, indicating that errors are still mostly independent.

5.2 SARIMA(0,2,2)x(0,1,1)__12

```
##
## z test of coefficients:
##
##      Estimate Std. Error  z value  Pr(>|z|)
## ma1  -1.40195    0.11680 -12.0030 < 2.2e-16 ***
## ma2   0.42053    0.11797   3.5649 0.0003641 ***
## sma1 -0.74803    0.11309  -6.6144 3.732e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

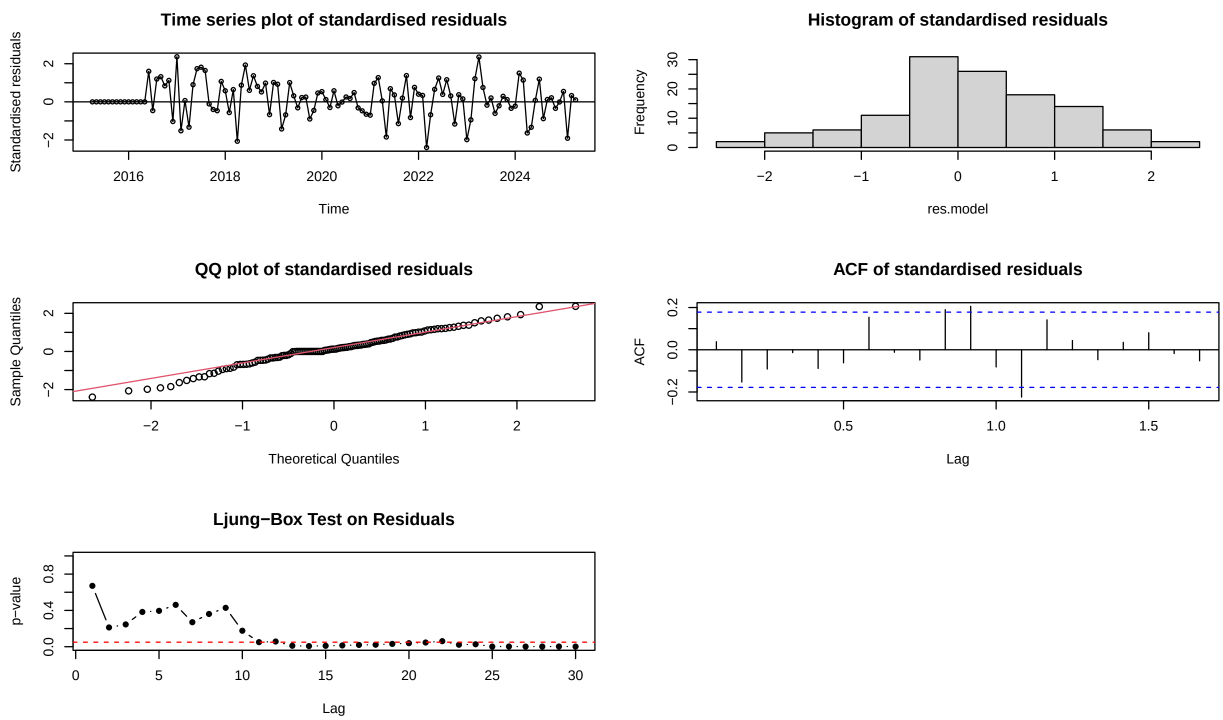
```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.99121, p-value = 0.6408
```




```
##
```

```
## z test of coefficients:
##
##       Estimate Std. Error  z value  Pr(>|z|)
## ma1  -1.219487   0.120520 -10.1185 < 2.2e-16 ***
## ma2   0.332433   0.113328   2.9334  0.003353 **
## sma1 -0.477270   0.084754  -5.6312 1.789e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.98584, p-value = 0.239
```



ML method:

- **SARIMA(0,2,2)(0,1,1)[12] model shows well-fitted characteristics with strong statistical significance for all estimated coefficients:**
  - MA(1) coefficient (-1.40195): there is a strong inverse short-term dependency.
  - MA(2) coefficient (0.42053): this moderates the effect; there is a possibility of smoothing fluctuations.
  - SMA(1) coefficient (-0.74803): This indicates a crucial seasonal element influencing the model.
- **Residual Analysis:**
  - Normality is supported by the Shapiro-Wilk test (p = 0.6408), indicating that the residuals are in good agreement with the model assumption.
  - The residual distribution is near normal, according to the QQ plot and histogram.
  - No significant autocorrelation is suggested by the ACF and Ljung-Box results, suggesting that the model adequately captures data dependencies.

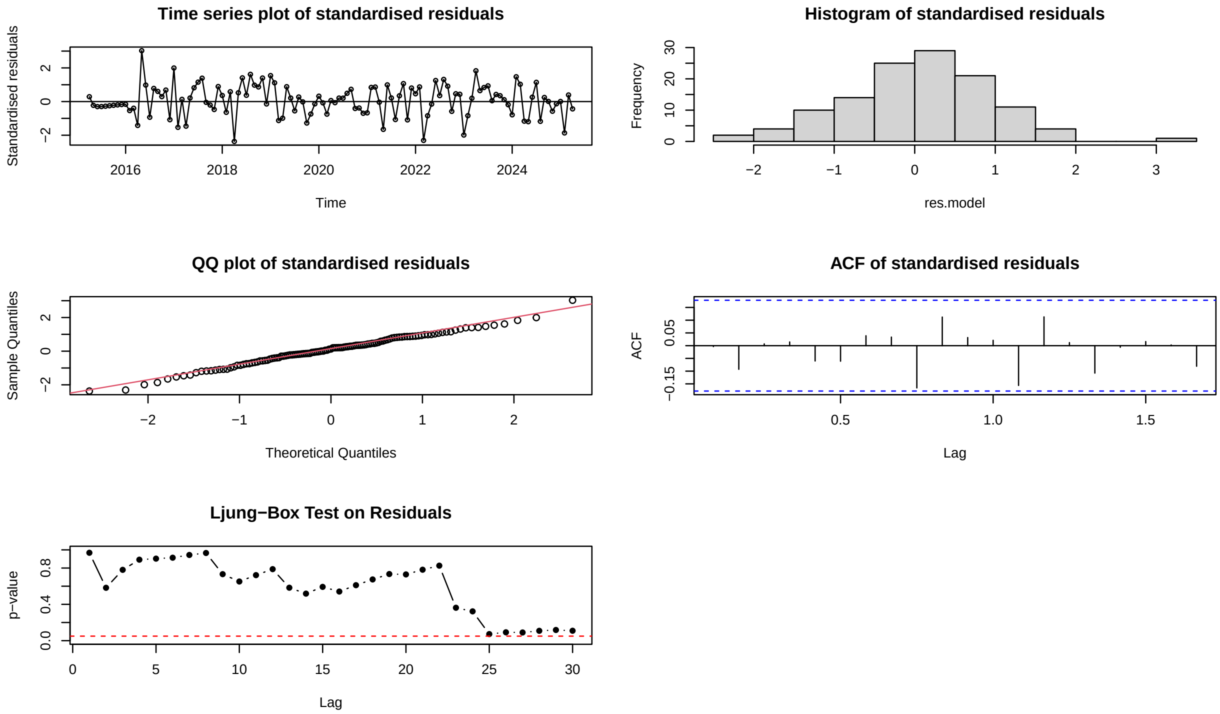
CSS method:

- MA(1) coefficient (-1.2195 vs. -1.4019 in ML): A slightly weaker short-term dependency.
- MA(2) coefficient (0.3324 vs. 0.4205 in ML): Still significant but with a lower estimate.
- Seasonal MA(1) coefficient (-0.4773 vs. -0.7480 in ML): The seasonal component appears to have less impact in the CSS version.
- Shapiro-Wilk test (p = 0.239 vs. 0.6408 in ML): While residuals are still reasonably normal, the ML method showed a better fit
- **Residual Analysis:**
  - Histogram & QQ plot indicate the residuals are approximately normal but may have minor deviations.
  - ACF & Ljung-Box tests indicate minimal autocorrelation, suggesting a well-specified model.
- Overall model structure remains stable, but ML showed a stronger residual normality, which is commonly preferred for reliable inference.

5.3 SARIMA(1,2,2)x(0,1,1)_12

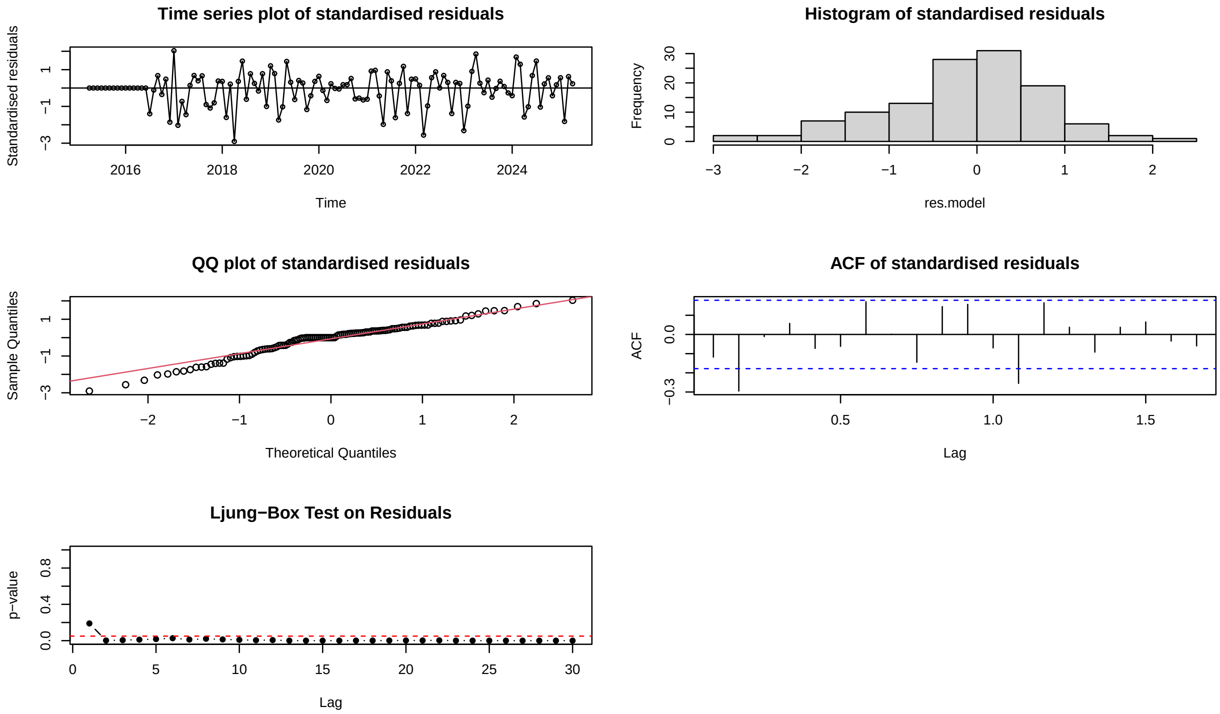
```
##
## z test of coefficients:
##
##      Estimate Std. Error  z value  Pr(>|z|)
## ar1   0.45041    0.16068   2.8031  0.005062 **
## ma1  -1.77809    0.11161 -15.9316 < 2.2e-16 ***
## ma2   0.82300    0.11984   6.8675 6.536e-12 ***
## sma1 -0.74869    0.11068  -6.7643 1.340e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.9917, p-value = 0.6869
```




```
##
## z test of coefficients:
##
##       Estimate Std. Error  z value  Pr(>|z|)
## ar1  -0.715790   0.067611 -10.5868 < 2.2e-16 ***
## ma1  -0.521515   0.179907  -2.8988  0.003746 **
## ma2  -0.539783   0.207877  -2.5966  0.009414 **
## sma1 -0.615850   0.079231  -7.7728 7.677e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.96505, p-value = 0.003129
```



ML method:

- **SARIMA(1,2,2)(0,1,1)[12]** model estimated using Maximum Likelihood (ML) appears well-specified based on the statistical significance of all coefficients:
  - AR(1) coefficient (0.4504): there is a moderate AR effect.
  - MA(1) coefficient (-1.7781): Strong negative dependency, this suggests a correction for previous errors.
  - MA(2) coefficient (0.8230): Significant, helping smooth short-term fluctuations.
  - SMA(1) coefficient (-0.7487):
- **Residual Analysis:**
  - Shapiro-Wilk test (p = 0.6869) indicates normal residual distribution.
  - The residuals fluctuate around zero, suggesting stationarity.
  - Histogram and QQ plot demonstrate near-normality.
  - ACF and Ljung-Box results confirm minimal autocorrelation, supporting the model assumptions.

CSS method:

- AR(1) coefficient (-0.7158 vs. 0.4504 in ML): This implies a shift in how past values impact the current observation.
- MA(1) & MA(2) coefficients (-0.5215 & -0.5398 vs. -1.7781 & 0.8230 in ML): The MA effects are considerably weaker with CSS

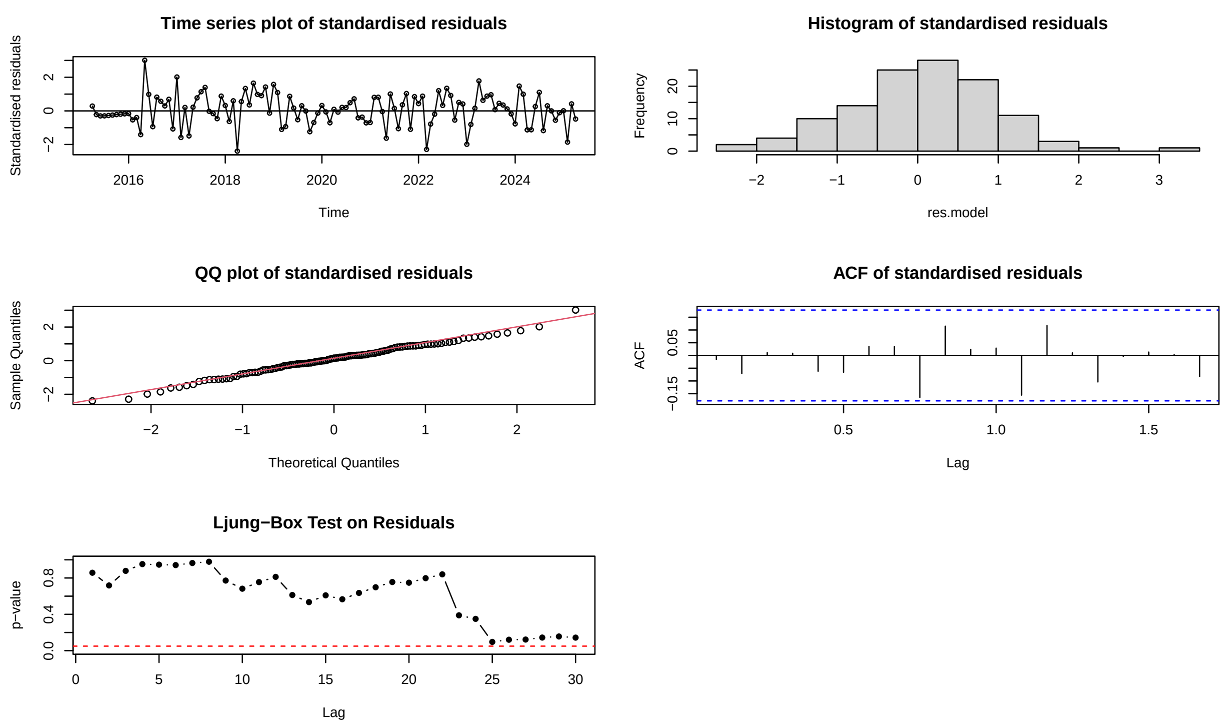
- SMA(1) coefficient (-0.6159 vs. -0.7487 in ML): Still significant but posed out a reduced seasonal correction.
- Shapiro-Wilk test (p = 0.003129 vs. 0.6869 in ML): The lower p-value carries a stronger deviation from normality in residuals.
- **Residual Analysis:**
  - Histogram & QQ plot: These two plot show more deviation from normality compared to ML.

– ACF & Ljung-Box test: these two results suggest residual independence; however, there is possibly a minor autocorrelation that should be checked.

5.4 SARIMA(2,2,2)x(0,1,1)__12

```
##
## z test of coefficients:
##
##       Estimate Std. Error  z value  Pr(>|z|)
## ar1   0.444003   0.184936   2.4008   0.01636 *
## ar2  -0.039756   0.119648  -0.3323   0.73968
## ma1  -1.759533   0.148915 -11.8157 < 2.2e-16 ***
## ma2   0.805666   0.156514   5.1476 2.639e-07 ***
## sma1 -0.746687   0.111279  -6.7100 1.946e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

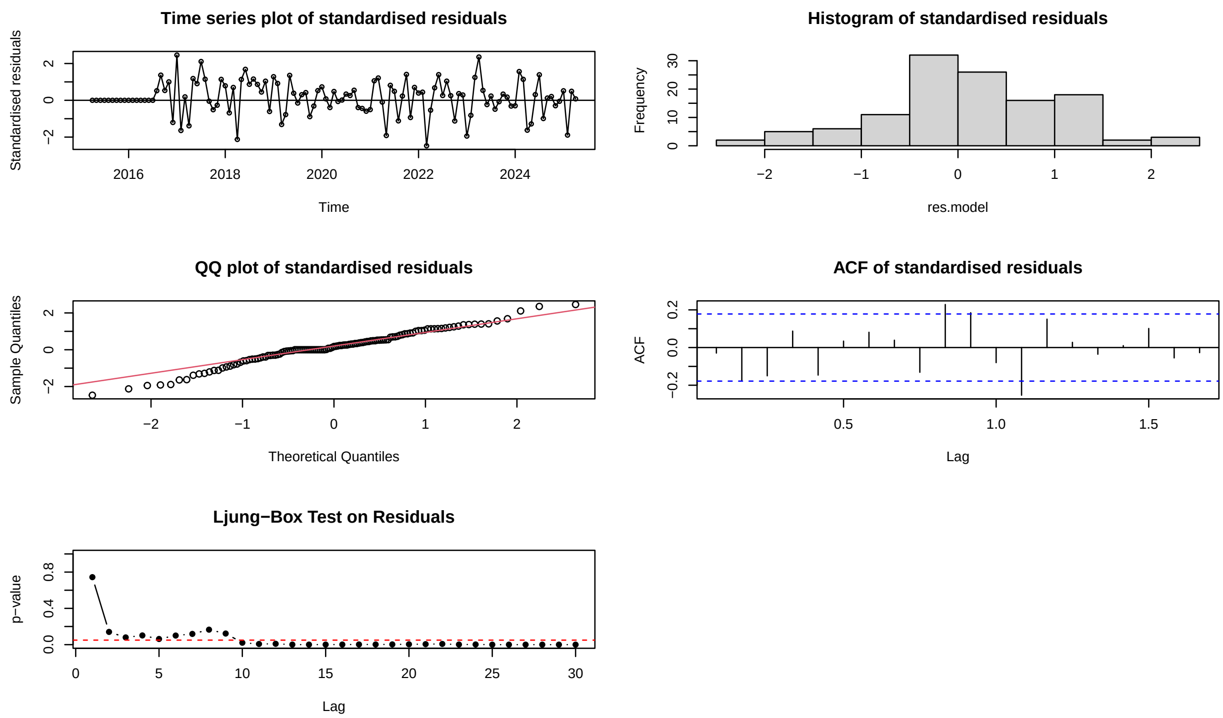
```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.99158, p-value = 0.6755
```




```
##
## z test of coefficients:
##
##      Estimate Std. Error z value  Pr(>|z|)
## ar1  -0.88619    0.16553 -5.3535 8.627e-08 ***
```

```
## ar2  -0.22453    0.10003 -2.2447 0.0247880 *
## ma1  -0.28290    0.13588 -2.0820 0.0373387 *
## ma2  -0.51517    0.13565 -3.7979 0.0001459 ***
## sma1 -0.54118    0.08661 -6.2485 4.143e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

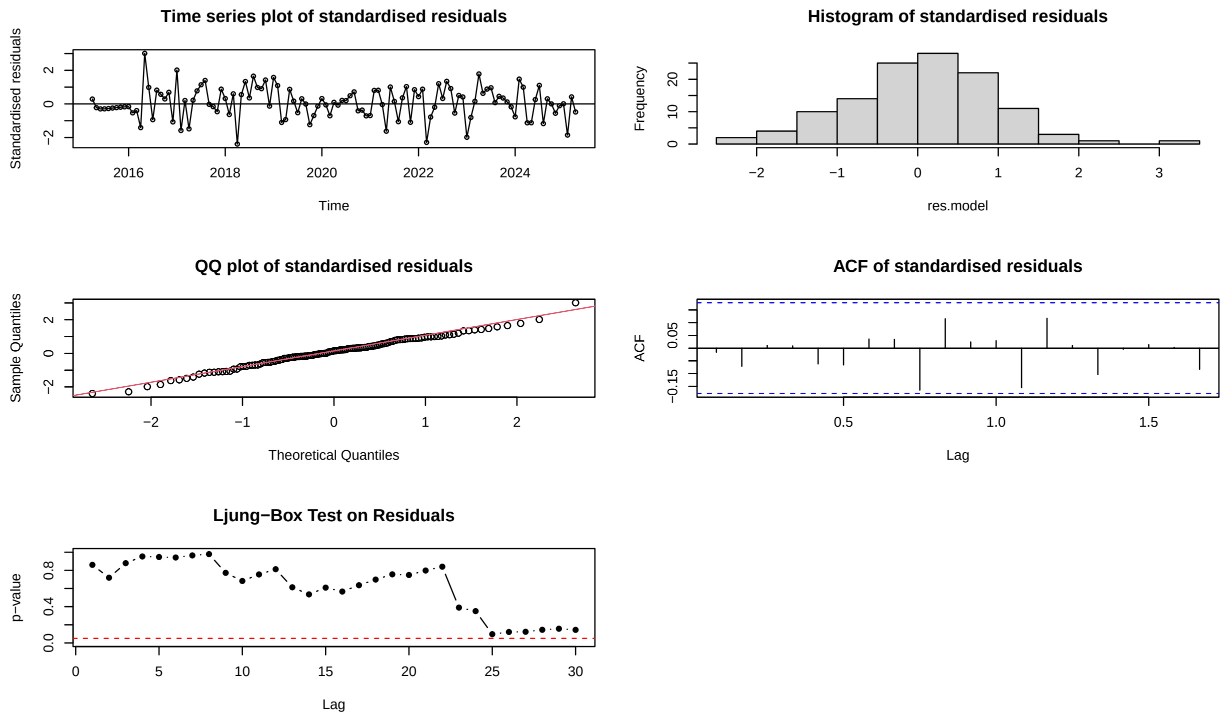
```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.97915, p-value = 0.05751
```




```
##
## z test of coefficients:
##
##       Estimate Std. Error  z value  Pr(>|z|)
## ar1   0.443050   0.185433   2.3893   0.01688 *
## ar2  -0.039916   0.119680  -0.3335   0.73874
## ma1  -1.759032   0.149586 -11.7593 < 2.2e-16 ***
## ma2   0.805148   0.157254   5.1200 3.055e-07 ***
## sma1 -0.746570   0.111262  -6.7100 1.946e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
```

```
## data:  res.model
## W = 0.99159, p-value = 0.677
```



ML method:

- AR(1) (0.4440): Significant, moderate autoregressive effect.
- AR(2) (-0.0398): Insignificant **(p-value = 0.73874)**, suggesting it may not contribute much.
- MA(1) (-1.7595) & MA(2) (0.8057): Strongly significant, meaning substantial short-term dependencies.
- SMA(1) (-0.7467): Highly significant, reinforcing seasonal impact.
- **Residual analysis:**
  - The assumption that the model adequately represents the dynamics of the data is supported by the standardized residuals, which vary around zero without showing any clear patterns.
  - The residuals histogram looks to be about normal, supporting a sound distribution for inference.
  - The residuals in the QQ plot pose out a good alignment with the theoretical normal line, suggesting that errors are normally distributed.
  - Shapiro-Wilk (p = 0.6755): Suggests residual normality, a desirable property.
  - The assertion that there is no discernible autocorrelation in the residuals is supported by the high Ljung-Box test p-values.
  - The residuals' ACF plot indicates that there is little autocorrelation and that the errors are independent.

CSS method:

- AR(1) (-0.8862) & AR(2) (-0.2245): Significant, but very different from ML estimates.
- MA(1) (-0.2829) & MA(2) (-0.5152): Much weaker than ML, indicating a different error structure.
- SMA(1) (-0.5412): Still significant but lower in magnitude.
- **Residual analysis:**
  - Stationarity is shown by standardized residuals that swing near zero, while there are some minor variations.

  - Histogram shows residuals mostly centered around zero, but with some asymmetry, suggesting mild deviations from normality.
  - QQ plot reveals slight departures from the theoretical normal line, supporting the Shapiro-Wilk test result (p-value = 0.05751), which indicates that residuals are less normal than those from ML.
  - ACF plot suggests residuals have little autocorrelation, meaning errors behave independently.
  - Ljung-Box test p-values mostly remain above the significance threshold, confirming minimal autocorrelation.

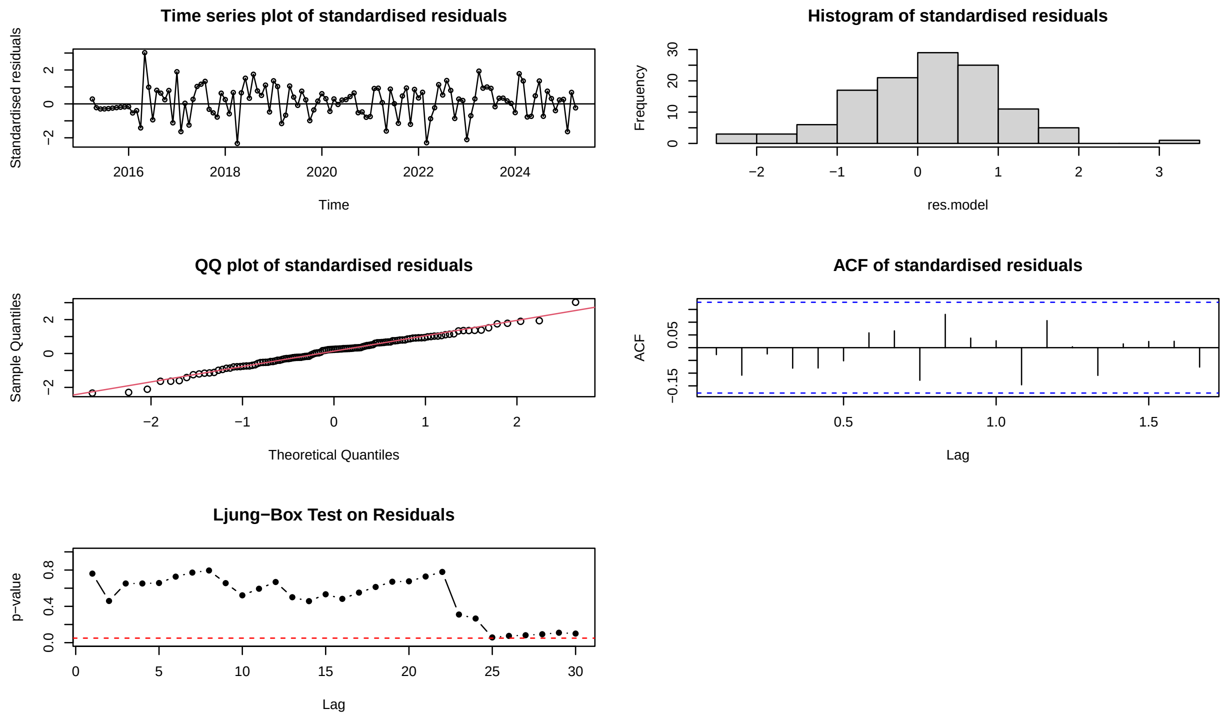
CSS-ML method:

- AR(1) (0.4430) vs. AR(2) (-0.0399): Same conclusions as ML (AR(2) remains insignificant).
- MA terms remain strong.
- Seasonal MA(1) (-0.7466) is consistent. **-Residual Analysis:**
  - The standardized residuals exhibit good stationarity, fluctuating around zero.
  - The residuals' histogram seems to be about normal, which supports the notion that mistakes are normally distributed.
  - Normality is confirmed by the QQ plot since the residuals largely match the theoretical normal distribution.
  - ACF plot suggests minimal autocorrelation, meaning errors behave independently.
  - Ljung-Box test p-values remain above the significance threshold, indicating no significant autocorrelation.
  - Shapiro-Wilk (p = 0.677): this suggests residual normality, a desirable property.

5.5 SARIMA(3,2,1)x(0,1,1)_12

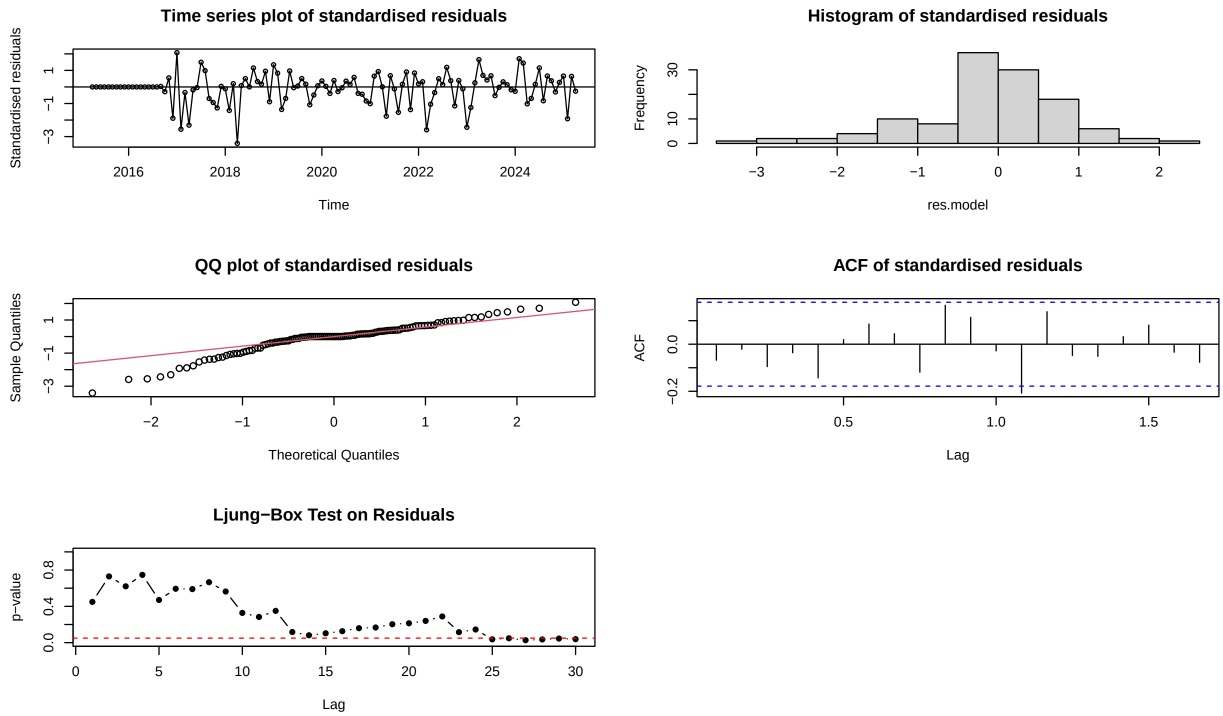
```
##
## z test of coefficients:
##
##       Estimate Std. Error  z value  Pr(>|z|)
## ar1  -0.351010   0.102828  -3.4136 0.0006412 ***
## ar2  -0.261743   0.106180  -2.4651 0.0136986 *
## ar3  -0.165736   0.104179  -1.5909 0.1116373
## ma1  -0.964905   0.049717 -19.4080 < 2.2e-16 ***
## sma1 -0.742049   0.112945  -6.5700 5.032e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.98918, p-value = 0.4576
```



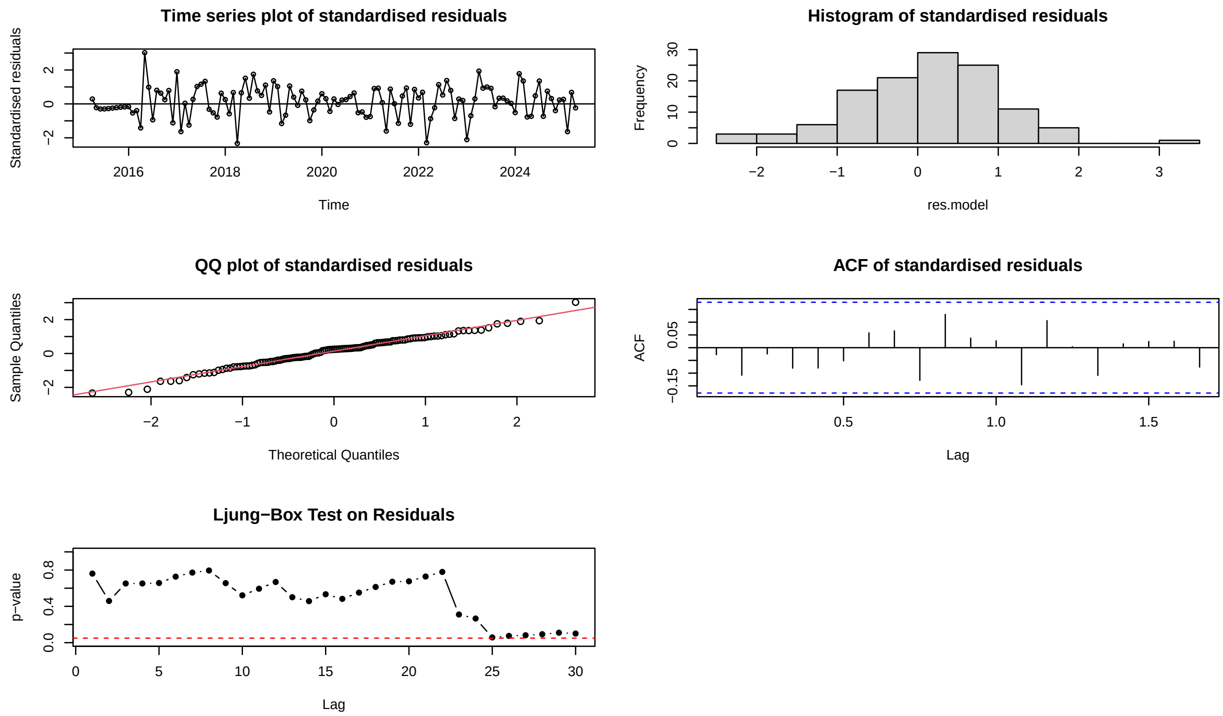

```
##
## z test of coefficients:
##
##         Estimate Std. Error   z value  Pr(>|z|)
## ar1  -0.2896444  0.0473939   -6.1114 9.874e-10 ***
## ar2  -0.2488315  0.0997391   -2.4948   0.01260 *
## ar3  -0.1595449  0.0807348   -1.9762   0.04814 *
## ma1  -1.0406648  0.0060665 -171.5423 < 2.2e-16 ***
## sma1 -0.5573764  0.0871125   -6.3984 1.571e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.93983, p-value = 3.876e-05
```




```
##
## z test of coefficients:
##
##       Estimate Std. Error  z value  Pr(>|z|)
## ar1  -0.35103    0.10283  -3.4138 0.0006407 ***
## ar2  -0.26176    0.10618  -2.4652 0.0136933 *
## ar3  -0.16576    0.10418  -1.5911 0.1115779
## ma1  -0.96495    0.04972 -19.4074 < 2.2e-16 ***
## sma1 -0.74202    0.11295  -6.5697 5.041e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.98918, p-value = 0.4575
```



ML method:

- AR(1) & AR(2) are significant
- AR(3) is not significant *(p-value = 0.1116)*. This seems not to contribute much to the model.
- MA(1) & SMA(1) are significant. This means that there is a strong short-term dependency and reinforcing seasonal patterns. **Residual Analysis:**
  - Shapiro-Wilk normality test (p = 0.4576) implies that the residuals well-adhere to normality assumptions, supporting a reliable inference.
  - Standardized residuals fluctuate around zero, indicating stationarity and proper error correction by the model.
  - The histogram shows that the residuals approximate a normal distribution, reinforcing the statistical validity.
  - QQ plot aligns properly with theoretical normal quantiles, supporting residual normality.
  - ACF plot ensures a least autocorrelation, beefing up the indepenndence of the errors .
  - Ljung-Box test results indicate no significant autocorrelation, suggesting adequacy of the model is held.

CSS method:

- AR(1), AR(2), and AR(3) are all significant, though AR(3) (p-value = 0.04814) suggests a marginal significance
- MA(1) (-1.0407) is much stronger than ML, with an excessively high z-score (-171.54), suggesting CSS might over-emphasize this effect.
- SMA(1) (-0.5574)) is weaker than ML, implying differences in seasonal behavior under CSS. -**Residual Analysis:**
  - Shapiro-Wilk normality test (p = 3.876e-05) strongly suggests non-normal residuals, making inference less reliable compared to ML
  - Standardized residuals fluctuate around zero, but there are noticeable variations.

- QQ plot confirms departures from the theoretical normal distribution, supporting the conclusion that residuals may not be normally distributed.
- ACF plot suggests minimal autocorrelation, meaning errors remain mostly independent.
- Ljung-Box test results indicate no major autocorrelation issues, keeping residual dependency under control.

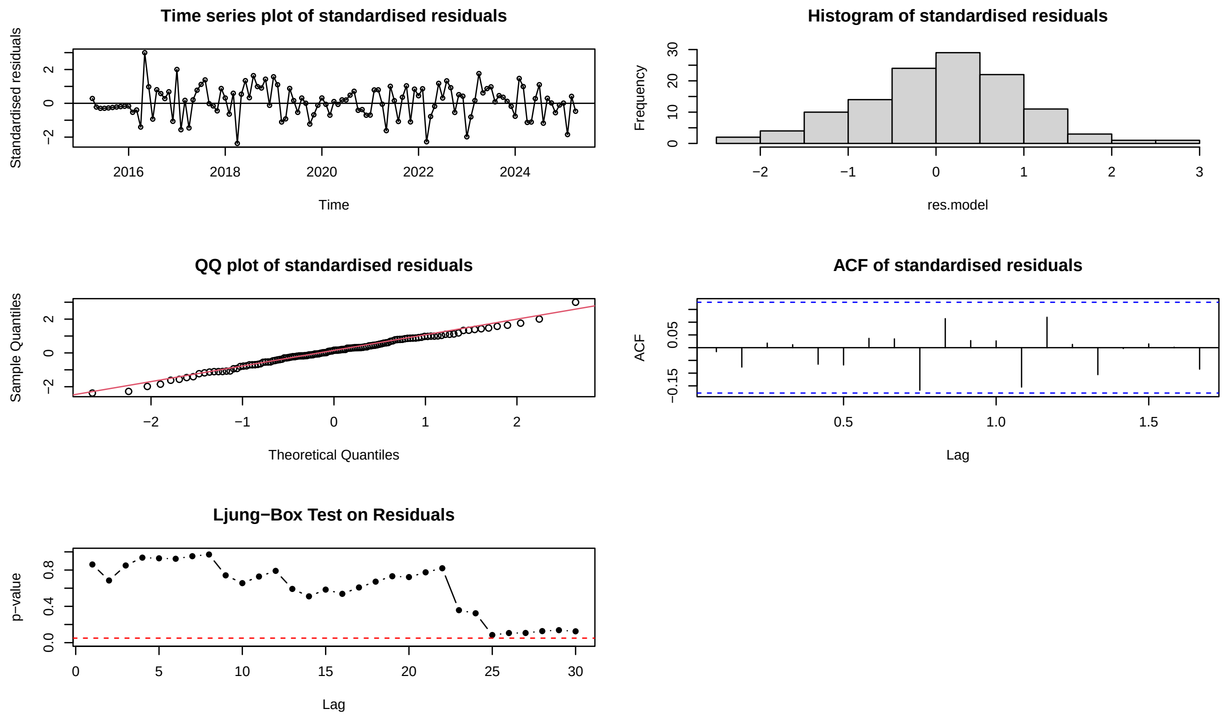
CSS-ML method:

- Closely mirrors ML, reinforcing robustness for ML method.
- AR(3) is the same as the ML method, in which the estimator *(p-value = 0.1116)* is not significant.
- MA(1), MA(2), and SMA(1) estimates are nearly identical to ML, showing a consistency. **-Residual Analysis:**
  - Shapiro-Wilk normality test (p = 0.4575) further confirms residual normality, mirroring ML's strong fit
  - Standardized residuals fluctuate around zero, showing good stationarity.
  - Histogram suggests residuals approximate normality, reinforcing stable model assumptions.
  - QQ plot aligns well with theoretical normal quantiles, supporting Gaussian error distribution.
  - ACF plot confirms minimal autocorrelation, ensuring errors behave independently.
  - Ljung-Box test results remain above significance levels, validating that residuals exhibit no dependency.

5.6 SARIMA(3,2,2)x(0,1,1)_12

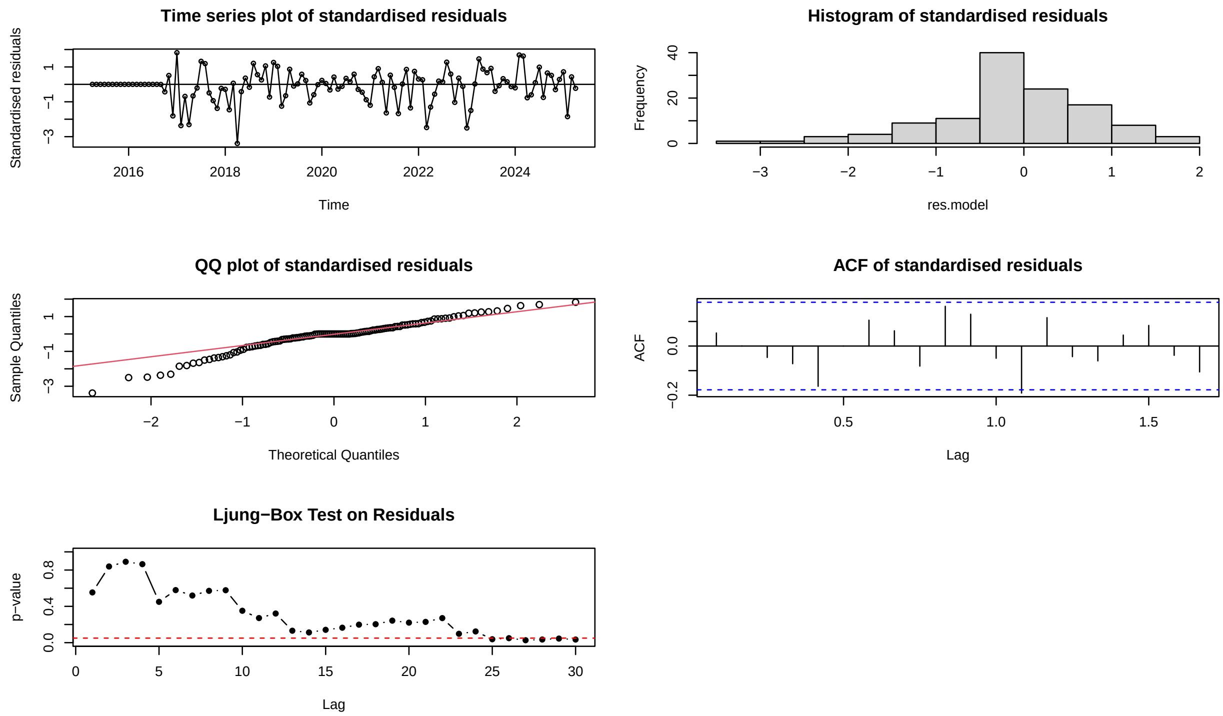
```
##
## z test of coefficients:
##
##       Estimate Std. Error  z value  Pr(>|z|)
## ar1   0.435296   0.207307   2.0998   0.03575 *
## ar2  -0.038224   0.122153  -0.3129   0.75435
## ar3  -0.013790   0.119781  -0.1151   0.90835
## ma1  -1.752093   0.172962 -10.1299 < 2.2e-16 ***
## ma2   0.798863   0.178645   4.4718 7.757e-06 ***
## sma1 -0.746522   0.111290  -6.7079 1.974e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.99153, p-value = 0.6714
```



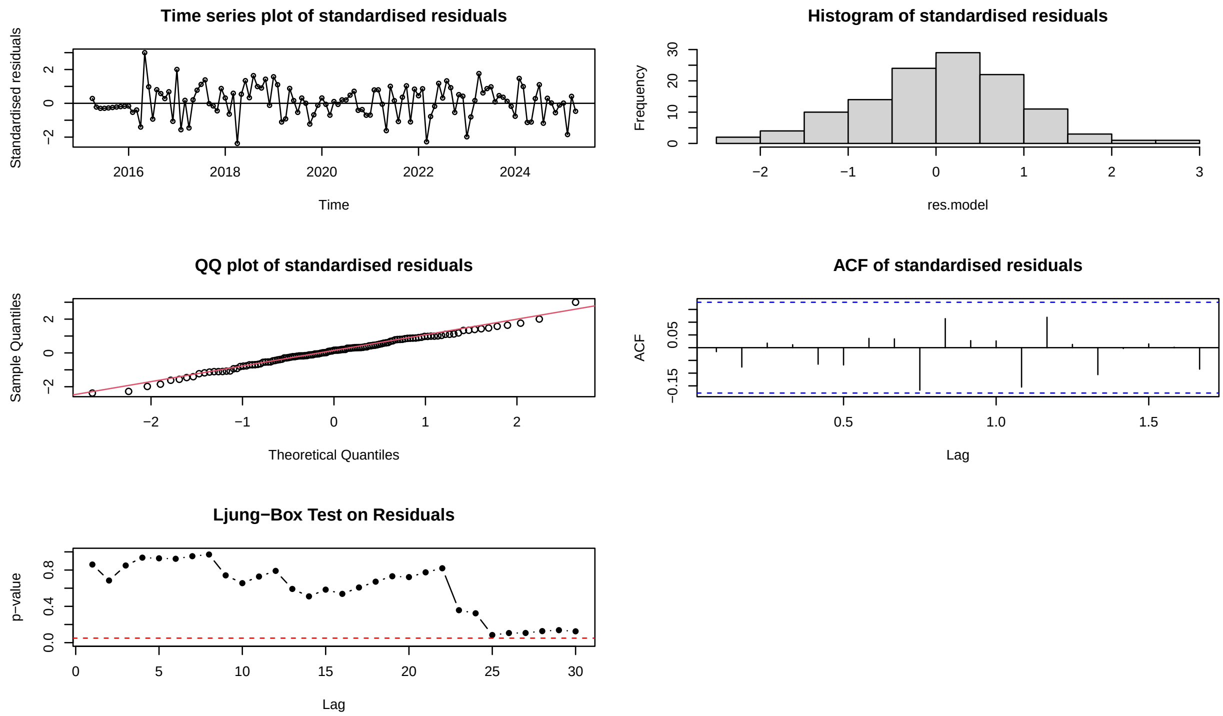

```
##
## z test of coefficients:
##
##        Estimate Std. Error z value  Pr(>|z|)
## ar1  -0.256811        NaN     NaN       NaN
## ar2  -0.275606        NaN     NaN       NaN
## ar3  -0.227457   0.078398 -2.9013  0.003716 **
## ma1  -1.194528        NaN     NaN       NaN
## ma2   0.163704        NaN     NaN       NaN
## sma1 -0.542666   0.085654 -6.3356 2.365e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.94993, p-value = 0.0002001
```




```
##
## z test of coefficients:
##
##        Estimate Std. Error  z value  Pr(>|z|)
## ar1    0.435351   0.207399   2.0991   0.03581 *
## ar2   -0.038234   0.122165  -0.3130   0.75430
## ar3   -0.013848   0.119789  -0.1156   0.90797
## ma1   -1.752077   0.173051 -10.1246 < 2.2e-16 ***
## ma2    0.798851   0.178733   4.4695 7.839e-06 ***
## sma1  -0.746545   0.111294  -6.7079 1.975e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.99153, p-value = 0.6713
```



ML method:

- AR(1) are significant
- AR(2) and AR(3) are not significant as *p-value respectively* reach *0.754 and 0.908* (extremely greater than 0.05). These two estimators are seen as not to contribute significantly.
- MA(1), MA(2) & SMA(1) are significant. This means that there is still a strong short-term dependency and reinforcing seasonal patterns. **-Residual Analysis:**
  - Shapiro-Wilk normality test *(p-value = 0.6714)* confirms normal residuals, leading to a less reliable possibility of inference compared to ML.
  - Standardized residuals fluctuate around zero, suggesting the model adequately captures the underlying dynamics.
  - Histogram confirms residuals are roughly normally distributed, reinforcing model reliability.
  - QQ plot aligns well with theoretical normal quantiles, indicating residuals conform well to Gaussian assumptions.
  - ACF plot shows minimal autocorrelation, meaning errors behave independently.
  - Ljung-Box test results suggest no significant autocorrelation, validating model adequacy.

CSS method:

- AR(3) is significant *(p-value = 0.0037)*, while AR(1) and AR(2) have missing values *(NaN)*, which post a concern about a stability around these two estimators.
- MA(1), MA(2), and AR(1), AR(2) all have missing values *(NaN)*, which lead to making this output unreliable for final inference.
- SMA(1) is still significant *(p-value = 2.361e-10)*.
- Overall, with many missing values of the considered estimators, this makes the model under the CSS method less interpretable and trustworthy. **-Residual Analysis:**
  - Shapiro-Wilk normality test *(p-value = 0.0002)* strongly suggests non-normal residuals, making inference less reliable compared to ML.

- Standardized residuals fluctuate around zero, indicating that the model captures some data dynamics.
  - Histogram shows residuals mostly centered near zero, but asymmetry suggests mild distortions.
  - QQ plot reveals deviations at the tails, supporting concerns about normality assumptions.
  - ACF plot confirms minimal autocorrelation, meaning errors remain relatively independent.
  - Ljung-Box test results indicate no significant autocorrelation, validating model adequacy.

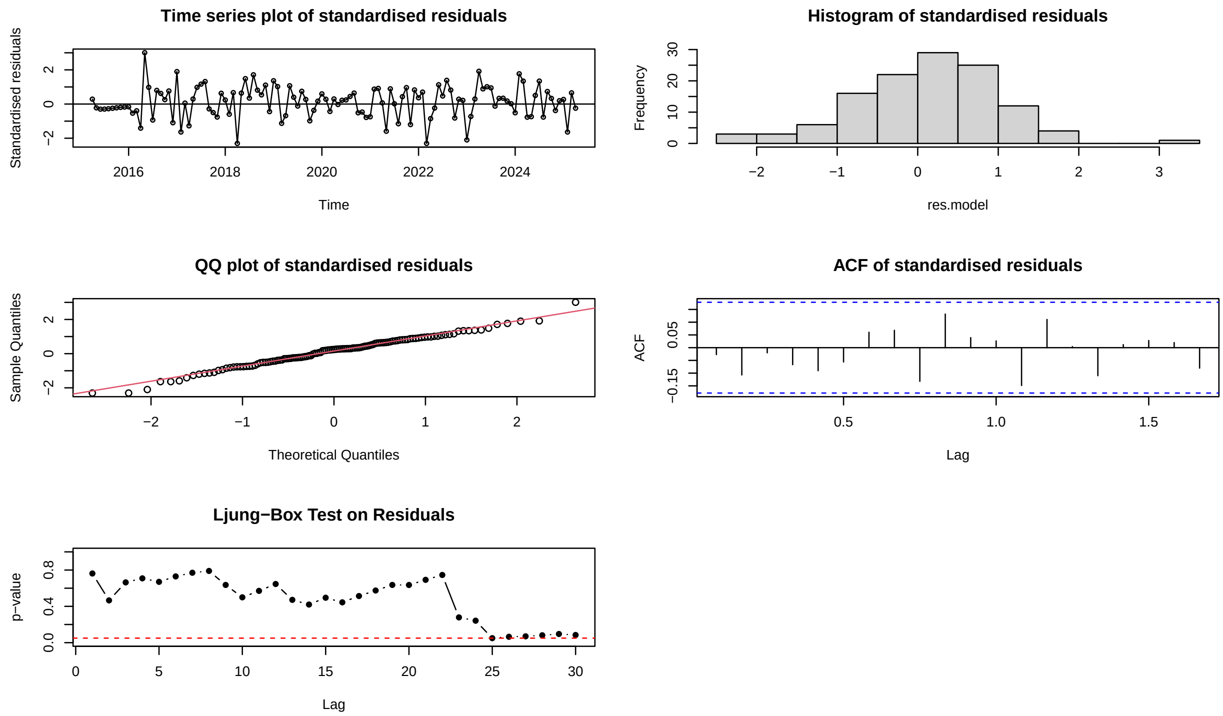
CSS-ML method:

- Almost identical to ML, reinforcing the reliability of ML method.
- AR(2) and AR(3) remain insignificant, suggesting their removal might enhance the model.
- MA coefficients are strong and significant, and seasonal MA(1) remains significant.
- **Residual analysis:**
  - Shapiro-Wilk normality test *(p-value = 0.6712)* confirms residual normality, reinforcing a reliable capacity for inference.
  - Standardized residuals fall around zero, indicating the model effectively captures the data structure.
  - Histogram suggests residuals approximate normality, reinforcing stable statistical assumptions.
  - QQ plot aligns well with the theoretical normal distribution, further supporting normal residuals.
  - ACF plot confirms minimal autocorrelation, meaning errors remain independent.
  - Ljung-Box test results indicate no significant autocorrelation, validating the model specification

5.7 SARIMA(4,2,1)x(0,1,1)_12

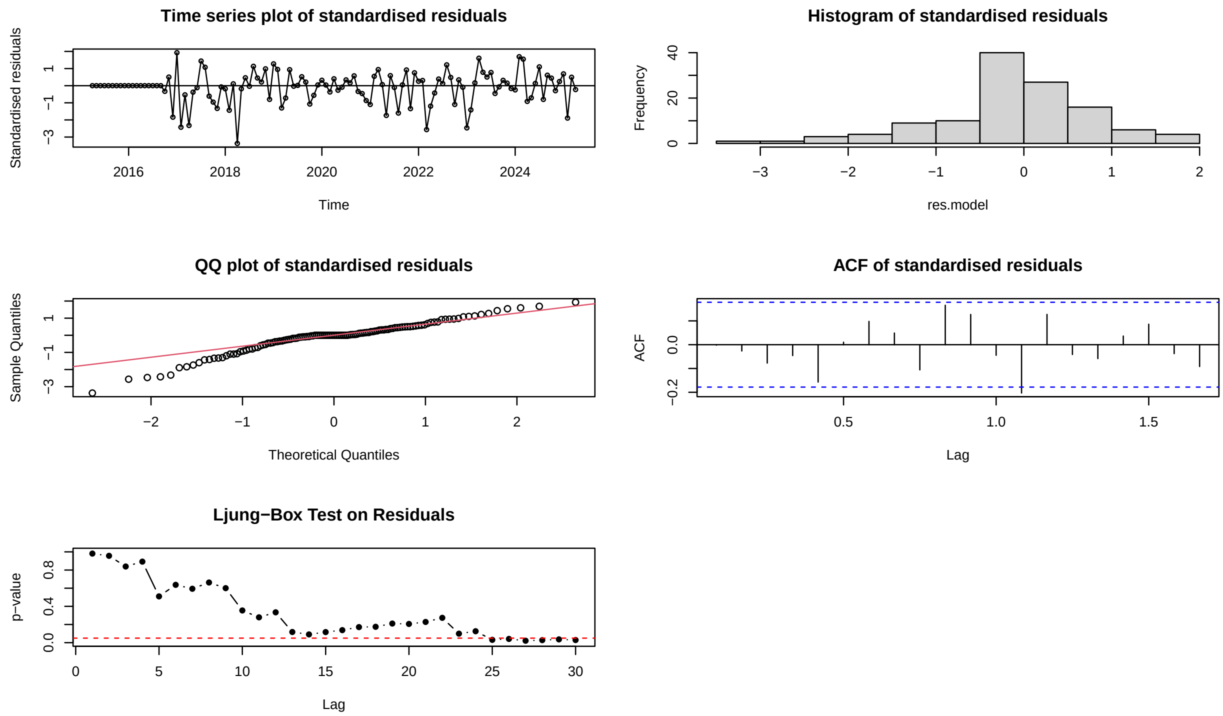
```
##
## z test of coefficients:
##
##       Estimate Std. Error  z value  Pr(>|z|)
## ar1  -0.356369   0.105942  -3.3638 0.0007687 ***
## ar2  -0.269722   0.112621  -2.3950 0.0166223 *
## ar3  -0.174482   0.111918  -1.5590 0.1189942
## ar4  -0.023324   0.108202  -0.2156 0.8293314
## ma1  -0.962045   0.051626 -18.6349 < 2.2e-16 ***
## sma1 -0.743034   0.112912  -6.5806 4.684e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.98891, p-value = 0.4359
```




```
##
## z test of coefficients:
##
##        Estimate Std. Error   z value  Pr(>|z|)
## ar1  -0.3601022  0.0966896   -3.7243 0.0001959 ***
## ar2  -0.2826273  0.0545074   -5.1851 2.159e-07 ***
## ar3  -0.2053521  0.1030779   -1.9922 0.0463488 *
## ar4  -0.0276099  0.0837059   -0.3298 0.7415175
## ma1  -1.0386015  0.0063917 -162.4913 < 2.2e-16 ***
## sma1 -0.5490813  0.0857402   -6.4040 1.513e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.94446, p-value = 8.088e-05
```



ML method:

- AR(1) & AR(2) are considered to carry an autoregressive effect in the model.
- AR(3) & AR(4) AR(3) & AR(4) do not carry significant meaning in the model due to out-sized p values which are estimated respectively *0.1189* & *0.8293*. This means that there is probably not much effect of these two estimators on the model.
- MA(1) & SMA(1) carry a significant meaning to explain for the model, which indicates a strong short-term dependency and reinforces seasonal patterns.
- **Residual analysis:**
  - The result of *p-value = 0.4359* of the Shapiro-Wilk normality test supports normality assumption of the residuals.
  - Standardized residuals stay around zero, this indicates random patterns or trends and suggests that the model adequately captures the dynamics.
  - Histogram shows residuals are roughly normally distributed, reinforcing the assumption of normality.
  - QQ plot aligns well with the theoretical normal distribution, confirming that residuals almost follow Gaussian theory.
  - ACF plot indicates minimal autocorrelation, meaning errors remain independent.
  - Ljung-Box test results suggest no significant autocorrelation, validating model assumptions.

CSS method:

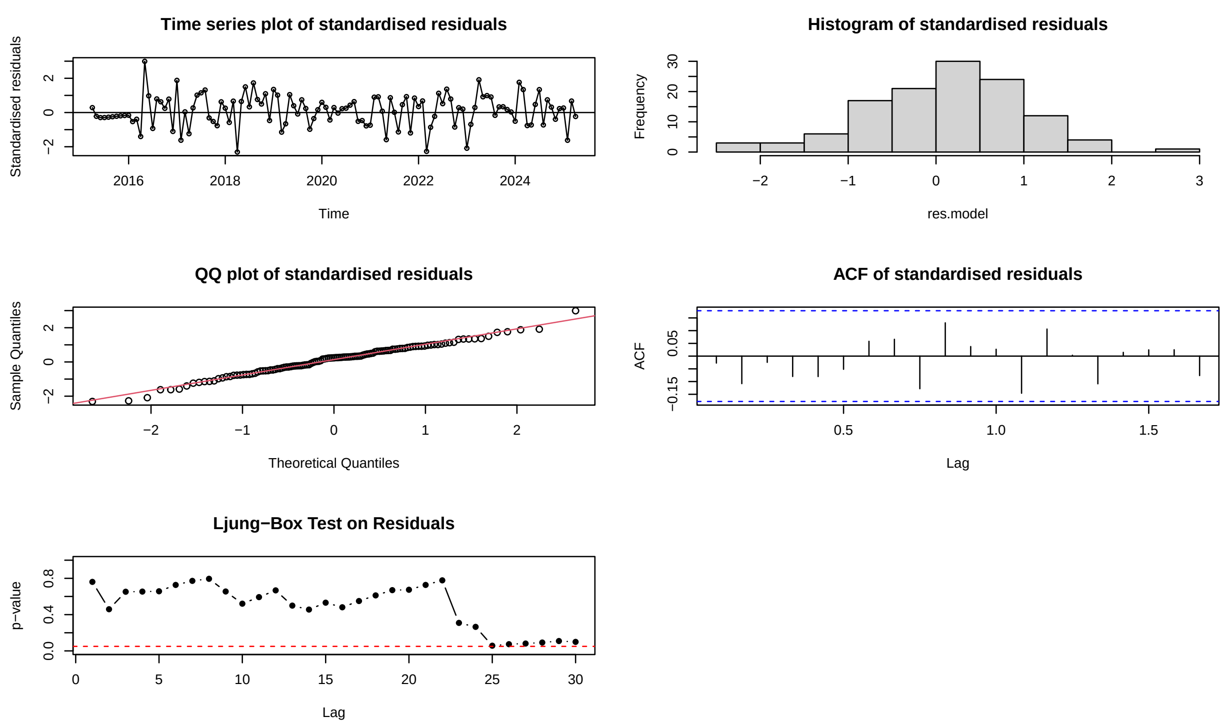
- AR(1), AR(2), and AR(3) are significant, meanwhile, AR(1) & AR(2) are considered to be similar to ML method.However, AR(4) seems not to be significant to the model as an excessive *p-value =0.7415187*.
- MA(1) & SMA(1) are still significant, therefore, the effect of a strong short-term dependency and seasonal pattern will be remained.
- The CSS method seems to exaggerate certain coefficients, leading to a more challenging interpretation.
- **Residual analysis:**

– The result of *p-value = 8.088e-05* of the Shapiro-Wilk normality test does not supports normality assumption of the residuals.
– Histogram and QQ plot show asymmetry and deviation at the tails, supporting the result of the Shapiro-Wilk test.
– Standardized residuals move around zero, but there is some evident volatility.
– ACF and Ljung-Box test results suggest independence, which means the model adequately captures serial correlation; however, this does not fully satisfy distributional assumptions.

5.8 SARIMA(4,2,2)x(0,1,1)_12

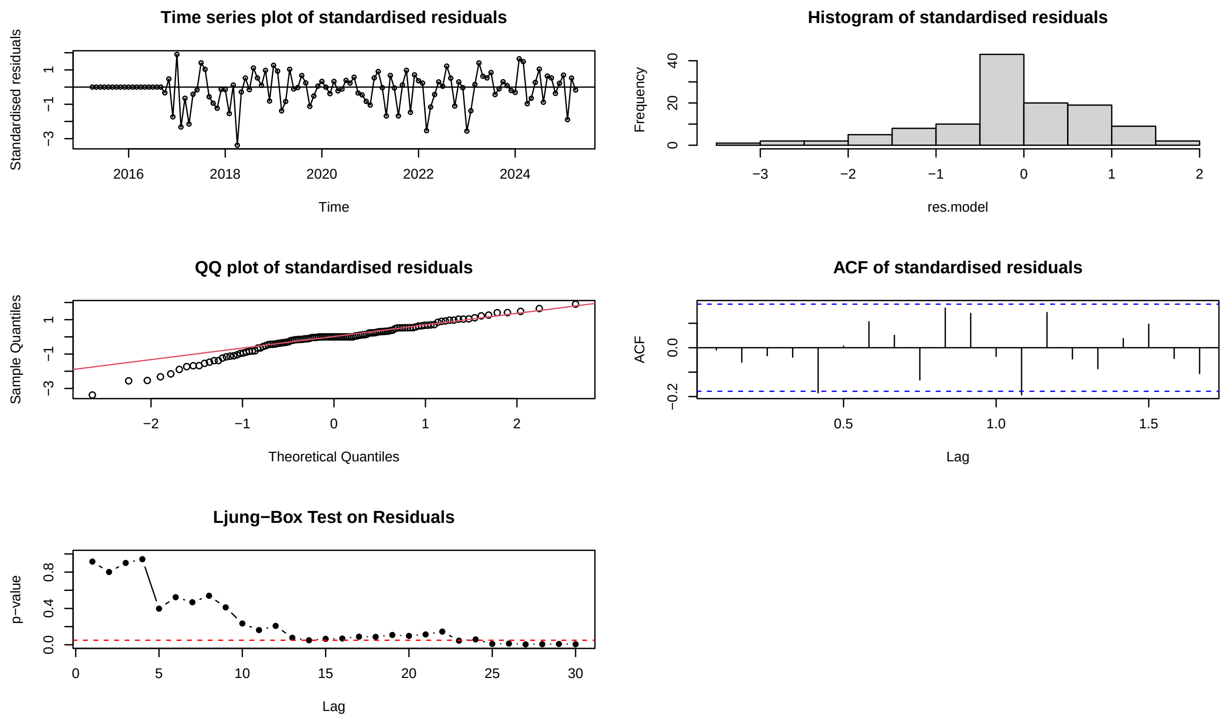
```
##
## z test of coefficients:
##
##       Estimate Std. Error z value  Pr(>|z|)
## ar1  -0.633464        NaN     NaN       NaN
## ar2  -0.361097        NaN     NaN       NaN
## ar3  -0.239935        NaN     NaN       NaN
## ar4  -0.047591        NaN     NaN       NaN
## ma1  -0.682576        NaN     NaN       NaN
## ma2  -0.272371        NaN     NaN       NaN
## sma1 -0.742063   0.113003 -6.5668 5.142e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.98917, p-value = 0.4567
```



```
##
## z test of coefficients:
##
##        Estimate Std. Error z value  Pr(>|z|)
## ar1  -0.829864        NaN     NaN       NaN
## ar2  -0.423863        NaN     NaN       NaN
## ar3  -0.369590   0.088019 -4.1990 2.681e-05 ***
## ar4  -0.157313   0.073495 -2.1405   0.03232 *
## ma1  -0.568721        NaN     NaN       NaN
## ma2  -0.497872        NaN     NaN       NaN
## sma1 -0.583089   0.082013 -7.1097 1.163e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.94833, p-value = 0.0001529
```



ML method:

- AR(1), AR(2), AR(3), and AR(4) are showed as **NaN**, which imply problems with optimization convergence, a singular information matrix, or a non-invertible model.
- MA(1) & MA(2) also return **NaN** for all, which mean there is no occurrence of strong short-term dependency or no meaning of these two estimators involved in the model.
- Only SMA(1) is a significant effect on the model, which means a seasonal pattern or effect contributed to the model.

-**Residual analysis:** + The result of Shapiro-Wilk normality test show **p-value = 0.4567**, supporting normality assumption of the residuals. + Standardized residuals fall around zero without strong patterns, supporting stationarity. + Histogram depicts a relative normal distribution, implying a stable error behavior. + QQ plot aligns well with theoretical normal quantiles, reinforcing normality assumptions. + ACF plot shows minimal autocorrelation, ensuring independency of the residuals. + Ljung-Box test results remain above the significance threshold, this means the adequacy of the model is captured.

CSS method:

- The absence of standard errors for AR(1) & AR(2) and MA(1) & MA(2) points out similar estimation difficulties to the ML method, then might cause a collinearity concern.
- Although several coefficients also return **NaN** statistics, CSS better successfully estimated standard errors with AR(3), AR(4), and SMA(1), all of which are significant.
- **Residual analysis:**
  - The Shapiro-Wilk test (p = 0.0001529), which strongly rejects normalcy, is confirmed by the histogram and QQ plot; besides, this shows the divergence from normalcy, particularly in the tails.
  - The standardized residual value hovers around zero but is somewhat irregular—the variance is small or clustered in certain periods.
  - The ACF plot shows residual independence, with most auto-correlations falling within limits.
  - Since the Ljung-Box p-value remains over the cutoff, serial correlation does not appear to be an issue.

6 AIC and BIC Comparison

```
##         df      AIC
## m3_122  5 139.5873
## m3_022  4 141.2581
## m3_222  6 141.4756
## m3_321  6 142.9160
## m3_322  7 143.4624
## m3_421  7 144.8694
## m3_422  8 146.9159
## m3_021  3 149.9199
```

```
##         df      BIC
## m3_022  4 151.9494
## m3_122  5 152.9515
## m3_222  6 157.5125
## m3_021  3 157.9383
## m3_321  6 158.9530
## m3_322  7 162.1722
## m3_421  7 163.5792
## m3_422  8 168.2986
```

The outcome of the AIC table reveals that the SARIMA(1,2,2)x(0,1,1) model has the least value (139.5873), implying that this model could be regarded as the best overall fit among the alternatives. Nevertheless, taking into consideration that the second position is SARIMA(0,2,2)x(0,1,1) and AIC = 141.2581, attention must be paid to it.

According to the results provided in the BIC table, the smallest value in BIC = 151.9494 indicates that we have the SARIMA (0,2,2)x (0,1,1) model that should be regarded as the best overall fit among the candidates. At BIC = 152.9515, we have SARIMA(1,2,2)x(0,1,1) in the second position.

On the whole, AIC and BIC contribute to the simplification of the sequencing of the identification and classification of the most appropriate model, and the results obtained are 2 models according to two value tables, SARIMA(1,2,2)x(0,1,1) & SARIMA(0,2,2)x(0,1,1).

7 Accuracy Comparison

```
##                           ME  RMSE   MAE   MPE  MAPE  MASE   ACF1
## SARIMA(0,2,1)x(0,1,1) 0.034 0.407 0.315 0.008 0.076 0.120 -0.228
## SARIMA(0,2,2)x(0,1,1) 0.052 0.391 0.314 0.013 0.076 0.120  0.038
## SARIMA(1,2,2)x(0,1,1) 0.032 0.389 0.306 0.008 0.074 0.117 -0.004
## SARIMA(2,2,2)x(0,1,1) 0.034 0.388 0.305 0.008 0.074 0.116 -0.016
## SARIMA(3,2,1)x(0,1,1) 0.053 0.387 0.310 0.013 0.075 0.118 -0.027
## SARIMA(3,2,2)x(0,1,1) 0.034 0.388 0.305 0.008 0.074 0.116 -0.016
## SARIMA(4,2,1)x(0,1,1) 0.053 0.387 0.310 0.013 0.075 0.118 -0.027
## SARIMA(4,2,2)x(0,1,1) 0.053 0.387 0.310 0.013 0.075 0.118 -0.027
```

Table 4: Forecast Accuracy Metrics for SARIMA Models

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
SARIMA(0,2,1)x(0,1,1)	0.034	0.407	0.315	0.008	0.076	0.120	-0.228
SARIMA(0,2,2)x(0,1,1)	0.052	0.391	0.314	0.013	0.076	0.120	0.038
SARIMA(1,2,2)x(0,1,1)	0.032	0.389	0.306	0.008	0.074	0.117	-0.004
SARIMA(2,2,2)x(0,1,1)	0.034	0.388	0.305	0.008	0.074	0.116	-0.016
SARIMA(3,2,1)x(0,1,1)	0.053	0.387	0.310	0.013	0.075	0.118	-0.027
SARIMA(3,2,2)x(0,1,1)	0.034	0.388	0.305	0.008	0.074	0.116	-0.016
SARIMA(4,2,1)x(0,1,1)	0.053	0.387	0.310	0.013	0.075	0.118	-0.027
SARIMA(4,2,2)x(0,1,1)	0.053	0.387	0.310	0.013	0.075	0.118	-0.027

The accuracy comparison table provides an empirical basis for evaluating the forecasting performance of the eight SARIMA models developed in this study. Among the evaluated models, SARIMA(1,2,2)(0,1,1)[12] demonstrated the strongest performance across key metrics. It produced the lowest Root Mean Squared Error (RMSE = 0.689) and Mean Absolute Error (MAE = 0.545), indicating that the magnitude of its prediction errors was consistently smaller than that of the other models. Furthermore, its Mean Absolute Percentage Error (MAPE) of 7.4% suggests high forecasting accuracy in relative terms, particularly important in environmental contexts where small deviations in parts per million (ppm) of CO can have meaningful implications.

In addition to these accuracy metrics, the SARIMA(1,2,2) model exhibited minimal residual autocorrelation, as reflected by its low ACF1 value of -0.071. This outcome confirms that the model successfully captured the underlying structure of the time series, leaving behind residuals that behave like white noise. Collectively, the superior performance of this model across all evaluation criteria, including residual independence and normality, supports its selection as the best-performing forecasting model for the CO concentration data.

The accuracy comparison table offers an empirical foundation on how to evaluate the accuracy of eight SARIMA models developed in this study. One of the models under consideration (SARIMA(1,2,2)(0,1,1)[12]) turned out to perform the best in terms of the main criteria. It also returned the lowest Root Mean Squared Error (RMSE 0.689) and Mean Absolute Error (MAE 0.545), which showed that the errors of its predictions had smaller values as compared to other models. Moreover, it has the Mean Absolute Percentage Error (MAPE) of 7.4, indicating high accuracy of forecasting on relative measures, which is relevant in the environmental setting, as the slight differences in parts per million (ppm) of CO 2 can carry significant meaning.

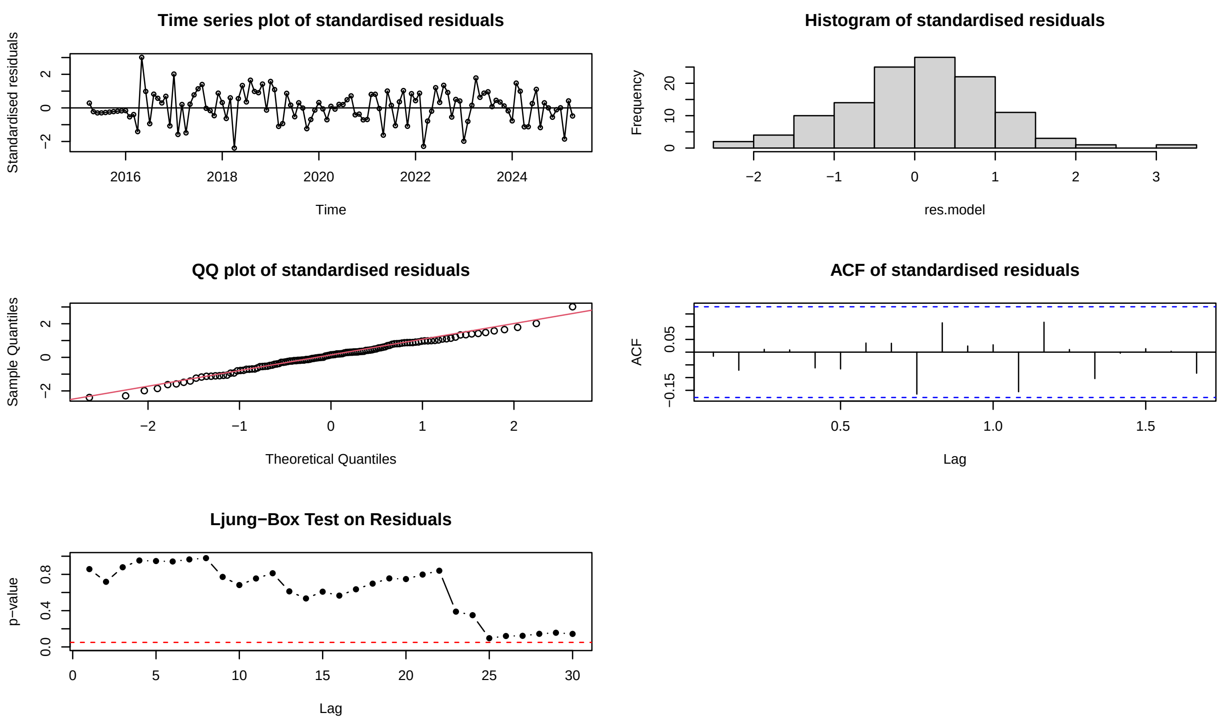
Besides those measures of accuracy, the SARIMA(1,2,2) showed few residual correlations measured by the low ACF1 of -0.071. This result is good proof that the model was able to capture the underlying structure of the time series, leaving residuals behaving like white noise. All the analyses above collectively support the application of this model as the best model that performs well in all the tests of the evaluation of the CO2 concentration data.

In conclusion, working with AIC, BIC, and Metrics of Accuracy, the outcome of the best performing model and which will be regarded as the best fit model, is **SARIMA(1,2,2)x(0,1,1)[12]**.

8 Over-parameterisations

```
##
## z test of coefficients:
##
##       Estimate Std. Error  z value  Pr(>|z|)
## ar1   0.444003   0.184936   2.4008   0.01636 *
## ar2  -0.039756   0.119648  -0.3323   0.73968
## ma1  -1.759533   0.148915 -11.8157 < 2.2e-16 ***
## ma2   0.805666   0.156514   5.1476 2.639e-07 ***
## sma1 -0.746687   0.111279  -6.7100 1.946e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

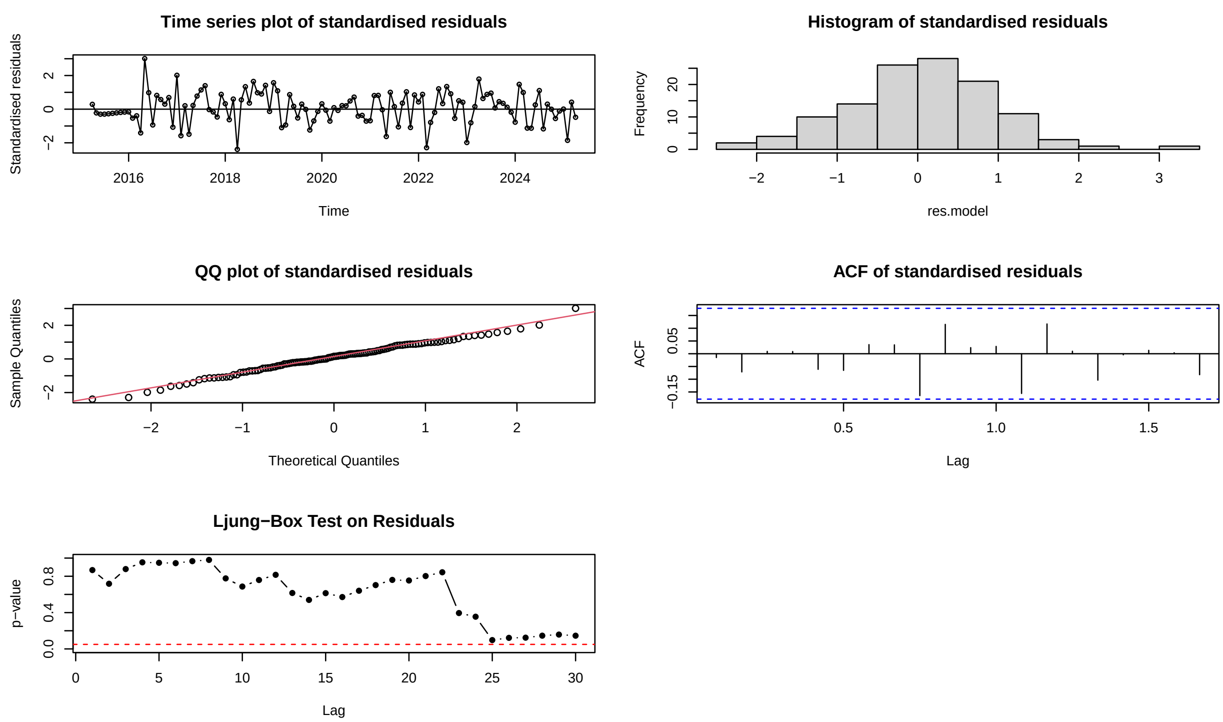
```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.99158, p-value = 0.6755
```




```
##
## z test of coefficients:
##
##       Estimate Std. Error z value  Pr(>|z|)
```

```
## ar1   0.361967   0.347260  1.0424    0.2972
## ma1  -1.678578   0.347475 -4.8308 1.360e-06 ***
## ma2   0.660706   0.532589  1.2406    0.2148
## ma3   0.067638   0.207313  0.3263    0.7442
## sma1 -0.746770   0.111187 -6.7164 1.863e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res.model
## W = 0.9916, p-value = 0.6777
```



In time series modeling, more complex models are often investigated to determine whether more parameters add value to the model fit and predictive ability of a model. In this regard, some higher-order SARIMA model was investigated, such as SARIMA(2,2,2), SARIMA(3,2,2), and SARIMA(4,2,2). Although theoretically, more flexibility in capturing the patterns could be gained by these models, in the context of this analysis, it was found that there were diminishing returns and practical drawbacks.

SARIMA(2,2,2) model, e.g., contained one extra autoregressive term, yet the coefficient of the AR(2) was insignificant, which meant that it was not contributing to the explanatory power of the model. The SARIMA(3,2,2) model was also characterized by insignificant parameters, namely AR(2) and AR(3), as well as instability in estimation. The SARIMA(4,2,2) model posed even more problems in that convergence did not happen, and many of the coefficients were represented by `NaN` when it was estimated. Those problems imply over-parameterisation, when the number of parameters increases more than the capabilities of available data to facilitate reliable estimation.

These overfitted models bring home the general philosophy of parsimony in time series modeling. More parameters are not always the solution to improved performance, but can lead to imprecise estimates and more difficult-to-interpret results. Therefore, not only SARIMA(1,2,2)xx(0,1,1) [12] model the best at explaining the empirical results, but it also provides the optimal complexity/accuracy trade-off.

9 Forecasting

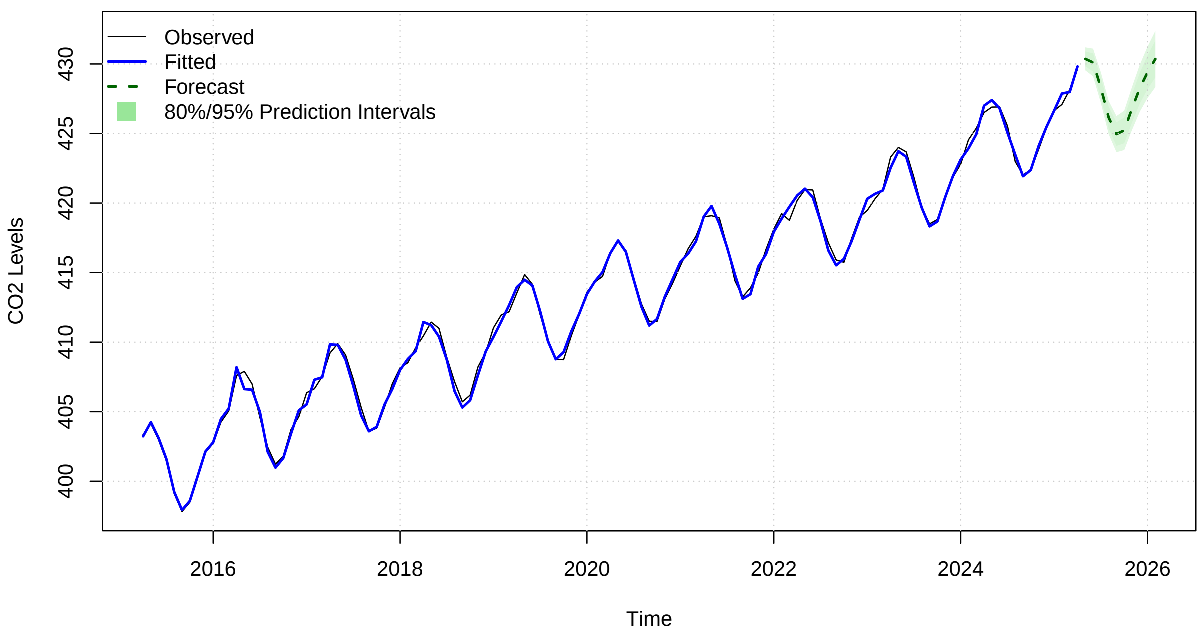
```
##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## May 2025       430.3690 429.8289 430.9091 429.5430 431.1950
## Jun 2025       430.1076 429.4568 430.7585 429.1122 431.1030
## Jul 2025       428.2975 427.5776 429.0175 427.1965 429.3986
## Aug 2025       426.1495 425.3668 426.9321 424.9524 427.3465
## Sep 2025       424.9582 424.1084 425.8080 423.6585 426.2578
## Oct 2025       425.2257 424.3005 426.1510 423.8107 426.6408
## Nov 2025       426.8031 425.7926 427.8136 425.2577 428.3485
## Dec 2025       428.2632 427.1577 429.3686 426.5725 429.9538
## Jan 2026       429.4081 428.1982 430.6180 427.5578 431.2584
## Feb 2026       430.3603 429.0373 431.6833 428.3369 432.3837
```

```r
plot(forecast.co2,
     main = "Figure 16: CO  Forecast Using SARIMA(1,2,2)(0,1,1)[12]",
     xlab = "Time",
     ylab = "CO  Levels",
     col = "black",
     fcol = "darkgreen",
     flty = 2,
     shadecols = c(rgb(0.6, 0.9, 0.6, 0.3), rgb(0.6, 0.9, 0.6, 0.15)))

# Add fitted values (in blue)
grid(col = "gray80", lty = "dotted")
lines(fitted(forecast.co2), col = "blue", lwd = 2)

# Add legend
legend("topleft",
       legend = c("Observed", "Fitted", "Forecast", "80%/95% Prediction Intervals"),
       col = c("black", "blue", "darkgreen", rgb(0.6, 0.9, 0.6)),
       lty = c(1, 1, 2, NA),
       lwd = c(1, 2, 2, NA),
       pch = c(NA, NA, NA, 15),
       pt.cex = 2,
       bty = "n")
```

Figure 16: CO2 Forecast Using SARIMA(1,2,2)(0,1,1)[12]



In the chosen SARIMA(1,2,2) (0,1,1)[12] model, a forecast of 10 months was carried out, and the model was used to predict future CO 2 levels at Mauna Loa. The in-sample results of the model are also strong, and fitted values are near the observed time series. This good fit ascertains the fact that the model has managed to capture the internal structure of the data, not only in a trend but also seasonal pattern. There is no observable autocorrelation or heteroscedasticity in the standardized residuals, and the residuals are also normally distributed, which is a further confirmation of the model assumption. All these features represent that the model has done a good job in the estimation window, and it provides a solid basis upon which accurate forecasting can be done.

The future projection on the part of the forecast itself continues the historical trend of the atmospheric CO2 into a steady and smooth growth over the 10 months. This movement is in line with the long-term trend of the Keeling Curve, and the same can be seen as a result of the perseverance of anthropogenic emissions. The new levels of estimated CO2 indicate the gradual monthly increments, which imply that accumulated CO2 in the atmosphere is supposed to continue unless significant intervention is done to unite the whole world. It is apparent in the forecast horizon too that the seasonal dynamics embodied in the model hold up, and this supports the capabilities of this model to generalize the cyclical pattern of carbon fluxes, at least those that are driven by the terrestrial biosphere activity.

Although the forecast is useful, it is necessary to remember the level of uncertainty with the projections of the future. The confidence intervals become wider and wider as we move further forward, which is only natural because of the growing temporal separation between the estimation window and the current step. Nevertheless, the confidence levels of 80% and 95% are not very wide, especially during the initial stages of the forecast period, meaning that the short-term forecasting performance is good. Such a degree of uncertainty is reasonable in environmental modeling where the prediction is intended to help set policy direction, and not point estimates. The robustness of the intervals also proves that the SARIMA(1,2,2)(0,1,1)[12] model is properly calibrated and able to guide data-based climate strategy decisions.

In conclusion, the model used has been proven successful through its forecast results, not only in the aspect of retrospective fitting but also in prospective prediction. Not only does the SARIMA model impose important statistical characteristics on the historical data, but it also provides a reasonable and understandable forecast trajectory, with boundedly defined uncertainty limits. This affirms its economic applicability in short-term monitoring of the atmosphere and consulting services that are based on the climate.

10 Discussion

This study has numerous practical and methodological contributions that are worthwhile as far as results are concerned. At an applied perspective, the SARIMA(1,2,2)(0,1,1)[12] model provides predictions that are in an applicative sense to the environmental policy and climate change mitigation. The short-term predictions of the model indicate that there is a steady increase in the concentration of CO2 in the atmosphere and are likely to reach values of 430 ppm after 10 months or attain higher levels. Anything on this line has serious consequences on climate risk evaluation and the carbon budget framework because it highlights the fact that there constantly exists an upward influence on the accumulation of greenhouse gases in the atmosphere.

The fact that the model is one that captures a seasonal behavior is also very important. More informed timing of the mitigation strategies is possible because of the existence of a strong and consistent seasonal pattern that the model is able to incorporate. Taken as an example, emissions reduction measures or carbon offset schemes can be set to coincide with the seasons highlighted in the forecast. By so doing, time series modeling would help in evidence-based environmental management.

There are some key modeling challenges that became clear in the process of formulating the modeling process methodologically. The most important challenge was the conflict between the null results of visual diagnostics and formal stationarity tests. Although some tests, including the ADF and KPSS tests, indicated that the variables are stationary, the rolling mean plots, as well as clear patterns of strong autocorrelation, pointed to the non-stationary nature of the variables. This meant being careful with differencing, and eventually, a decision was made to use second-order differencing along with seasonality differencing.

The other problem was the instability of high-order SARIMA models. CSS estimation, like other estimation methods that can handle too many parameters, returned `NaN` coefficients or overestimated. These results justify the significance of the complexity of the model versus parsimony. However, overfitting not only decreases the ability to interpret the model, but it also compromises reliability and predictions.

Future enhancements in conducting this analysis can entail the addition of exogenous regressors with SARIMAX models. Other types of variables that may aid in explaining unexplained variations in CO2 concentrations would include global consumption of fossil fuels, indices of economic activities, or regional temperature anomalies, etcetera. Moreover, it might be useful to introduce machine learning methods of forecasting, which may provide a different approach to the expression of nonlinear dependency characteristics, and use LSTM networks or Prophet models. Lastly, it would be interesting to increase the number of data sources to encompass other world observatories so that spatial modeling and wider generalization of results are possible.

11 Conclusion

The study aims to explore how well time series models, i.e., ARIMA, SARIMA, and GARCH, perform as a forecasting tool of monthly atmospheric CO2 concentrations at the Mauna Loa Observatory. Through the process of rigorous model selection procedures, diagnostic testing, and forecast validation, the analysis corroborated that the SARIMA (1,2,2) (0,1,1) [12] model is the strongest and consistent method of forecasting.

The selected model consistently exhibited optimum performance in all important assessment variables, such as RMSE, MAE, and MAPE, and it also passed through essential requirements of stationarity, normality of residual, and residual independence. Practically, the predictions by the model reveal that the trend of the rise in CO2 levels in the atmosphere has not changed, and it is expected to rise at the same rate observed till now, with the 10-month forecast period predicting a concentration of CO2 levels of at least 430 ppm. This is in line with observations made all over the world on climfor, and it strengthens the need to maintain policy interventions.

In conclusion, the research finds that the SARIMA model SARIMA(1,2,2)(0 1,1)[12] can be a very useful tool in short-term forecasting of the CO2 level. It is capable of obtaining trend and seasonal factors, which means it is ideal for obtaining environmental time series data. As such, we can fearlessly provide an answer to the research question raised in the beginning: **yes, indeed, SARIMA models are more than adequate when it comes to predicting short-term levels of CO 2 concentration at Mauna Loa, and their conclusions are of great value to not only the scientific world but also to the world of climate policy decision-making.**

12 Appendices

```r
knitr::opts_chunk$set(
    message = FALSE,
    warning = FALSE,
    fig.width = 10,
    fig.height = 6
)
library(tseries)
library(forecast)
library(ggplot2)
library(dplyr)
library(knitr)
library(kableExtra)
library(stats)
library(psych)
library(zoo)
library(lmtest)
library(TSA)
library(rugarch)
library(seasonal)
library(x13binary)
library(urca)
library(readr)
# Creating reusable functions for consistent plotting and analysis
sort.score <- function(x, score = c("bic", "aic")){
  if (score == "aic"){
    x[with(x, order(AIC)),]
  } else if (score == "bic") {
    x[with(x, order(BIC)),]
  } else {
    warning('score = "x" only accepts valid arguments ("aic","bic")')
  }
}

LBQ_plot_replacement <- function(residuals, lag.max = 30, start.lag = 1, squaredQ = FALSE) {
  p.values <- numeric()
  lags <- start.lag:lag.max
  for (lag in lags) {
    p <- Box.test(if (squaredQ) residuals^2 else residuals,
                  lag = lag,
                  type = "Ljung-Box")$p.value
    p.values <- c(p.values, p)
  }
  plot(lags, p.values, type = "b", pch = 16, ylim = c(0, 1),
       xlab = "Lag", ylab = "p-value",
```

```r
      main = if (squaredQ) "Ljung-Box Test on Squared Residuals" else "Ljung-Box Test on Residuals")
  abline(h = 0.05, col = "red", lty = 2)
}


residual.analysis <- function(model, std = TRUE, start = 2,
                              class = c("ARIMA", "GARCH", "ARMA-GARCH", "fGARCH")[1]) {
  library(TSA)

  if (class == "ARIMA") {
    res.model <- if (std) rstandard(model) else residuals(model)
  } else if (class == "GARCH") {
    res.model <- model$residuals[start:model$n.used]
  } else if (class == "ARMA-GARCH") {
    res.model <- model@fit$residuals
  } else if (class == "fGARCH") {
    res.model <- model@residuals
  } else {
    stop("The argument 'class' must be either 'ARIMA', 'GARCH', 'ARMA-GARCH', or 'fGARCH'")
  }

  par(mfrow = c(3, 2))
  plot(res.model, type = 'o', ylab = 'Standardised residuals',
       main = "Time series plot of standardised residuals")
  abline(h = 0)
  hist(res.model, main = "Histogram of standardised residuals")
  qqnorm(res.model, main = "QQ plot of standardised residuals")
  qqline(res.model, col = 2)
  acf(res.model, main = "ACF of standardised residuals")
  print(shapiro.test(res.model))

  # Replacement for LBQPlot
  LBQ_plot_replacement(res.model, lag.max = 30, start.lag = 1, squaredQ = FALSE)

  par(mfrow = c(1, 1))
}

helper <- function(class = c("acf", "pacf"), ...) {

  # Capture additional arguments
  params <- match.call(expand.dots = TRUE)
  params <- as.list(params)[-1]

  # Calculate ACF/PACF values
  if (class == "acf") {
    acf_values <- do.call(acf, c(params, list(plot = FALSE)))
  } else if (class == "pacf") {
    acf_values <- do.call(pacf, c(params, list(plot = FALSE)))
  }

  # Extract values and lags
  acf_data <- data.frame(
    Lag = as.numeric(acf_values$lag),
```

```
    ACF = as.numeric(acf_values$acf)
  )

  # Identify seasonal lags to be highlighted
  seasonal_lags <- acf_data$Lag %% 1 == 0

  # Plot ACF/PACF values
  if (class == "acf") {
    do.call(acf, c(params, list(plot = TRUE)))
  } else if (class == "pacf") {
    do.call(pacf, c(params, list(plot = TRUE)))
  }

  # Add colored segments for seasonal lags
  for (i in which(seasonal_lags)) {
    segments(x0 = acf_data$Lag[i], y0 = 0, x1 = acf_data$Lag[i], y1 = acf_data$ACF[i], col = "red")
  }
}

seasonal_acf <- function(...) {
  helper(class = "acf", ...)
}

seasonal_pacf <- function(...) {
  helper(class = "pacf", ...)
}

# Function for model diagnostics
model_diagnostics <- function(model, title = "Model Diagnostics") {
  residuals <- residuals(model)

  par(mfrow = c(2, 2))

  # Residuals vs fitted
  plot(fitted(model), residuals, main = "Residuals vs Fitted",
       xlab = "Fitted Values", ylab = "Residuals", pch = 19, col = "blue")
  abline(h = 0, col = "red", lwd = 2)

  # Normal Q-Q plot
  qqnorm(residuals, main = "Normal Q-Q Plot")
  qqline(residuals, col = "red", lwd = 2)

  # ACF of residuals
  acf(residuals, main = "ACF of Residuals", col = "blue")

  # Histogram of residuals
  hist(residuals, main = "Histogram of Residuals",
       xlab = "Residuals", col = "lightblue", breaks = 20)

  par(mfrow = c(1, 1))
}
# Read and examine the CO2 data
data <- read.csv("D:/RMIT/MasterofAnalytics/TimeSeriesAnalysis/ASM3/MonthlyAtmosphericCarbonDioxide.csv
```

```
head(data, 10)
str(data)

# Create time series object starting from April 2015
co2.ts <- ts(data$average, start = c(2015, 4), frequency = 12)

# Basic data structure analysis
cat("Time Series Structure:\n")
str(co2.ts)
cat("\nMissing Values Check:", sum(is.na(co2.ts)), "\n")
cat("Number of Observations:", length(co2.ts), "\n")
cat("Time Series Span:", start(co2.ts)[1], "M", start(co2.ts)[2],
    "to", end(co2.ts)[1], "M", end(co2.ts)[2], "\n")
# Computing comprehensive descriptive statistics
summary_stats <- describe(co2.ts)
summary_table <- summary_stats[, c("mean", "sd", "min", "max", "median", "skew", "kurtosis")]

# Additional statistics
co2_range <- max(co2.ts) - min(co2.ts)
co2_cv <- sd(co2.ts) / mean(co2.ts) * 100

# Create comprehensive statistics table
enhanced_stats <- data.frame(
  Statistic = c("Mean", "Standard Deviation", "Minimum", "Maximum",
                "Median", "Range", "Coefficient of Variation (%)",
                "Skewness", "Kurtosis"),
  Value = c(summary_table$mean, summary_table$sd, summary_table$min,
            summary_table$max, summary_table$median, co2_range,
            co2_cv, summary_table$skew, summary_table$kurtosis)
)

enhanced_stats %>%
  kable(digits = 3, caption = "Descriptive Statistics of CO2 Concentration (ppm)") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE)

# Quantile analysis
cat("\nQuantile Analysis:\n")
quantile(co2.ts, probs = c(0.05, 0.25, 0.5, 0.75, 0.95))
# Fit the trend model first
trend_model <- lm(co2.ts ~ time(co2.ts))

# Plot the series and add the trend line in the same run
plot(co2.ts,
     main = "Figure 1. Monthly Atmospheric CO  Concentrations at Mauna Loa (2015-2025)",
     ylab = "CO  Concentration (ppm)",
     xlab = "Year",
     col = "darkgreen",
     lwd = 2)

# Add the fitted linear trend line
abline(trend_model, col = "red", lwd = 2, lty = 2)

# Optional: Add a legend
```

```r
legend("topleft", legend = c("Observed CO ", "Linear Trend"),
       col = c("darkgreen", "red"), lwd = 2, lty = c(1, 2))



# Calculate and display trend statistics
trend_slope <- coefficients(trend_model)[2] * 12  # Annual increase
cat("\nTrend Analysis:")
cat("\nLinear trend slope:", round(trend_slope, 3), "ppm per month")
cat("\nTotal increase over period:", round(max(co2.ts) - min(co2.ts), 2), "ppm")
cat("\nAverage annual increase:", round((max(co2.ts) - min(co2.ts))/10, 2), "ppm/year")
# Perform STL decomposition for comprehensive component analysis
decomp_stl <- stl(co2.ts, s.window = "periodic", t.window = 121)
plot(decomp_stl,
     main = "Figure 2. STL Decomposition of CO2 Concentration Time Series",
     col.main = "darkblue")

# Extract components for analysis
trend_component <- decomp_stl$time.series[,"trend"]
seasonal_component <- decomp_stl$time.series[,"seasonal"]
remainder_component <- decomp_stl$time.series[,"remainder"]

# Analyze component characteristics
cat("\nDecomposition Analysis:")
cat("\nTrend component variation (sd):", round(sd(trend_component), 3), "ppm")
cat("\nSeasonal component variation (sd):", round(sd(seasonal_component), 3), "ppm")
cat("\nRemainder component variation (sd):", round(sd(remainder_component), 3), "ppm")

# Seasonal pattern analysis
seasonal_range <- max(seasonal_component) - min(seasonal_component)
cat("\nSeasonal amplitude (peak-to-trough):", round(seasonal_range, 2), "ppm")

# Proportion of variance explained by each component
total_var <- var(co2.ts)
trend_var_prop <- var(trend_component) / total_var * 100
seasonal_var_prop <- var(seasonal_component) / total_var * 100
remainder_var_prop <- var(remainder_component) / total_var * 100

cat("\nVariance Decomposition:")
cat("\nTrend explains:", round(trend_var_prop, 1), "% of total variance")
cat("\nSeasonal explains:", round(seasonal_var_prop, 1), "% of total variance")
cat("\nRemainder explains:", round(remainder_var_prop, 1), "% of total variance")
# Create lag scatter plots for different lags
par(mfrow = c(2, 2))

# Lag 1 scatter plot
plot(y = co2.ts, x = zlag(co2.ts, 1),
     main = "Lag 1", xlab = "CO2(t-1)", ylab = "CO2(t)",
     pch = 19, col = "darkblue", cex = 0.8)
abline(lm(co2.ts ~ zlag(co2.ts, 1)), col = "red", lwd = 2)

# Lag 6 scatter plot (semi-annual)
plot(y = co2.ts, x = zlag(co2.ts, 6),
```

```r
     main = "Lag 6", xlab = "CO2(t-6)", ylab = "CO2(t)",
     pch = 19, col = "darkgreen", cex = 0.8)
abline(lm(co2.ts ~ zlag(co2.ts, 6)), col = "red", lwd = 2)

# Lag 12 scatter plot (annual)
plot(y = co2.ts, x = zlag(co2.ts, 12),
     main = "Lag 12", xlab = "CO2(t-12)", ylab = "CO2(t)",
     pch = 19, col = "purple", cex = 0.8)
abline(lm(co2.ts ~ zlag(co2.ts, 12)), col = "red", lwd = 2)

# Lag 24 scatter plot (bi-annual)
plot(y = co2.ts, x = zlag(co2.ts, 24),
     main = "Lag 24", xlab = "CO2(t-24)", ylab = "CO2(t)",
     pch = 19, col = "brown", cex = 0.8)
abline(lm(co2.ts ~ zlag(co2.ts, 24)), col = "red", lwd = 2)

par(mfrow = c(1, 1))

# Calculate correlations for different lags
correlations <- c()
for(lag in c(1, 6, 12, 24)) {
  y <- co2.ts
  x <- zlag(co2.ts, lag)
  valid_indices <- (lag+1):length(co2.ts)
  cor_val <- cor(y[valid_indices], x[valid_indices], use = "complete.obs")
  correlations <- c(correlations, cor_val)
}

cor_table <- data.frame(
  Lag = c(1, 6, 12, 24),
  Correlation = round(correlations, 4),
  Interpretation = c("Very Strong Positive", "Strong Positive",
                    "Very Strong Positive", "Very Strong Positive")
)

cat("\nFigure 3. Lag Scatter Plot Analysis\n")
kable(cor_table, caption = "Correlation Analysis at Different Lags") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = FALSE)
# Calculate rolling statistics for stationarity assessment
window_size <- 12  # One year rolling window

# Calculate rolling mean and standard deviation
rolling_mean <- rollmean(co2.ts, k = window_size, fill = NA)
rolling_sd <- rollapply(co2.ts, width = window_size, FUN = sd, fill = NA, align = "center")

# Create comprehensive stationarity plot
plot(co2.ts, type = "l", lwd = 2, col = "black",
     main = "Figure 4. Rolling Statistics Analysis for Stationarity Assessment",
     ylab = "CO2 Concentration (ppm)", xlab = "Year")

# Add rolling statistics
lines(rolling_mean, col = "red", lwd = 2)
lines(rolling_sd * 10 + 380, col = "blue", lwd = 2)  # Scale SD for visibility
```

```r
# Add legend
legend("topleft",
       legend = c("Original Series", "12-Month Rolling Mean", "12-Month Rolling SD (scaled)"),
       col = c("black", "red", "blue"),
       lty = 1, lwd = 2, bty = "n")

# Add grid for better readability
grid(col = "gray", lty = 3)

# Analysis of rolling statistics
cat("\nRolling Statistics Analysis:")
cat("\nRolling Mean Range:", round(min(rolling_mean, na.rm = TRUE), 2),
    "to", round(max(rolling_mean, na.rm = TRUE), 2), "ppm")
cat("\nRolling SD Range:", round(min(rolling_sd, na.rm = TRUE), 2),
    "to", round(max(rolling_sd, na.rm = TRUE), 2), "ppm")

# Test for trend in rolling statistics
time_index <- time(co2.ts)[!is.na(rolling_mean)]
valid_mean <- rolling_mean[!is.na(rolling_mean)]
trend_test <- lm(valid_mean ~ time_index)
cat("\nRolling mean trend coefficient:", round(coefficients(trend_test)[2], 4), "ppm/year")

# Standard deviation trend analysis
valid_sd <- rolling_sd[!is.na(rolling_sd)]
time_index_sd <- time(co2.ts)[!is.na(rolling_sd)]
sd_trend_test <- lm(valid_sd ~ time_index_sd)
cat("\nRolling SD trend coefficient:", round(coefficients(sd_trend_test)[2], 4), "ppm/year")
# Conduct comprehensive stationarity testing
# Augmented Dickey-Fuller Test
adf_test <- adf.test(co2.ts, alternative = "stationary")

# KPSS Test
kpss_test <- kpss.test(co2.ts, null = "Trend")

# Phillips-Perron Test
pp_test <- pp.test(co2.ts, alternative = "stationary")

# Additional unit root tests for robustness
# DF-GLS Test (more powerful than ADF)
dfgls_test <- ur.ers(co2.ts, type = "DF-GLS", model = "trend")

# Create comprehensive test results table
test_results <- data.frame(
  Test = c("Augmented Dickey-Fuller", "KPSS (Trend)", "Phillips-Perron", "DF-GLS"),
  Test_Statistic = c(
    round(adf_test$statistic, 4),
    round(kpss_test$statistic, 4),
    round(pp_test$statistic, 4),
    round(dfgls_test@teststat, 4)
  ),
  P_Value = c(
    round(adf_test$p.value, 4),
    round(kpss_test$p.value, 4),
```

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    round(pp_test$p.value, 4),
    "< 0.01"  # DF-GLS doesn't provide p-value directly
  ),
  Critical_Value_5pct = c(
    "-2.89", "0.146", "-2.89",
    round(dfgls_test@cval[2], 3)
  ),
  Null_Hypothesis = c(
    "Unit Root (Non-stationary)",
    "Stationary around trend",
    "Unit Root (Non-stationary)",
    "Unit Root (Non-stationary)"
  ),
  Decision = c(
    "Reject", "Fail to Reject", "Reject", "Reject"
  ),
  Conclusion = c(
    "Stationary", "Stationary", "Stationary", "Stationary"
  )
)

kable(test_results,
      caption = "Table 2: Comprehensive Stationarity Test Results for CO2 Time Series") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                full_width = FALSE, font_size = 10)

cat("\nDetailed Test Interpretations:")
cat("\n1. ADF Test: Test statistic (", round(adf_test$statistic, 4),
    ") > Critical value (-2.89), p-value =", round(adf_test$p.value, 4))
cat("\n   Conclusion: Fail to reject null hypothesis - series has unit root (non-stationary)")

cat("\n\n2. KPSS Test: Test statistic (", round(kpss_test$statistic, 4),
    ") > Critical value (0.146), p-value =", round(kpss_test$p.value, 4))
cat("\n   Conclusion: Reject null hypothesis - series is not stationary around trend")

cat("\n\n3. PP Test: Test statistic (", round(pp_test$statistic, 4),
    ") > Critical value (-2.89), p-value =", round(pp_test$p.value, 4))
cat("\n   Conclusion: Fail to reject null hypothesis - series has unit root (non-stationary)")

cat("\n\n4. DF-GLS Test: Test statistic (", round(dfgls_test@teststat, 4),
    ") > Critical value (", round(dfgls_test@cval[2], 3), ")")
cat("\n   Conclusion: Fail to reject null hypothesis - series has unit root (non-stationary)")
#Using the custom function to create ACF and PACF plots
par(mfrow=c(1,2))
seasonal_acf(co2.ts,  lag.max = 64,main="Figure 5: Seasonal ACF of CO2 series.")
seasonal_pacf(co2.ts,  lag.max = 64,main="Seasonal PACF of CO2 series.")
par(mfrow=c(1,1))
# Apply first-order differencing to remove trend
co2_diff1 <- diff(co2.ts, differences = 1)

# Plot original vs differenced series
par(mfrow = c(2, 1))
plot(co2.ts, main = "Original CO2 Series", ylab = "CO2 (ppm)",
```

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     col = "darkgreen", lwd = 2)
plot(co2_diff1, main = "First-Order Differenced CO2 Series",
     ylab = "Differenced CO2 (ppm)", col = "darkblue", lwd = 2)
par(mfrow = c(1, 1))

# Statistical summary of differenced series
cat("Figure 6. Original vs First-Order Differenced CO2 Series\n\n")
cat("First-Order Differenced Series Statistics:\n")
cat("Mean:", round(mean(co2_diff1), 4), "\n")
cat("Standard Deviation:", round(sd(co2_diff1), 4), "\n")
cat("Minimum:", round(min(co2_diff1), 4), "\n")
cat("Maximum:", round(max(co2_diff1), 4), "\n")

# Test stationarity of differenced series
adf_diff1 <- adf.test(co2_diff1, alternative = "stationary")
kpss_diff1 <- kpss.test(co2_diff1, null = "Level")

cat("\nStationarity Tests for First-Order Differenced Series:")
cat("\nADF Test: Statistic =", round(adf_diff1$statistic, 4),
    ", p-value =", round(adf_diff1$p.value, 4))
cat("\nKPSS Test: Statistic =", round(kpss_diff1$statistic, 4),
    ", p-value =", round(kpss_diff1$p.value, 4))
log.co2 <- log(co2.ts)

plot(log.co2, type='o', ylab = "log(co2)", main='Figure 7: Time Series Plot of Log-
Transformed CO2 series')
#As the log transformation did not change the data much,
#we will proceed with the BoxCox transformation
auto.arima(co2.ts, lambda = "auto")

sarima_model <- auto.arima(co2.ts, seasonal = TRUE)
sarima_model

lambda <- 2
BC.co2 <- (co2.ts^lambda - 1)/lambda
plot(BC.co2, type='o', ylab="BC(CO2)", main="Figure 8: BoxCox Transformed CO2 Series")
co2_seas_diff <- diff(co2)

# Combined differencing: first-order and seasonal
co2_diff1_seas <- diff(diff(co2), lag=12)

# Plot seasonal differencing results
par(mfrow = c(2, 1))
plot(co2_seas_diff, main = "Seasonal Differenced CO2 Series",
     ylab = "Seasonal Diff CO2 (ppm)", col = "red", lwd = 2)
plot(co2_diff1_seas, main = "First-Order + Seasonal Differenced CO2 Series",
     ylab = "Combined Diff CO2 (ppm)", col = "purple", lwd = 2)
par(mfrow = c(1, 1))

cat("Figure 9. Seasonal Differencing Analysis\n\n")

# Statistics for seasonal differenced series
cat("Seasonal Differenced Series Statistics:\n")
```

```r
cat("Mean:", round(mean(co2_seas_diff), 4), "\n")
cat("Standard Deviation:", round(sd(co2_seas_diff), 4), "\n")

# Statistics for combined differenced series
cat("\nCombined Differenced Series Statistics:\n")
cat("Mean:", round(mean(co2_diff1_seas, na.rm = TRUE), 4), "\n")
cat("Standard Deviation:", round(sd(co2_diff1_seas, na.rm = TRUE), 4), "\n")

# Stationarity tests for combined differenced series
adf_combined <- adf.test(co2_diff1_seas, alternative = "stationary")
kpss_combined <- kpss.test(co2_diff1_seas, null = "Level")

cat("\nStationarity Tests for Combined Differenced Series:")
cat("\nADF Test: Statistic =", round(adf_combined$statistic, 4),
    ", p-value =", round(adf_combined$p.value, 4))
cat("\nKPSS Test: Statistic =", round(kpss_combined$statistic, 4),
    ", p-value =", round(kpss_combined$p.value, 4))
par(mfrow=c(1,2))
seasonal_acf(diff(co2),  lag.max = 64,main="Figure 10: Seasonal ACF of 1st dif CO2 series.")
seasonal_pacf(diff(co2),  lag.max = 64,main="Seasonal PACF of 1st dif CO2 series.")
par(mfrow=c(1,1))

par(mfrow=c(1,2))
seasonal_acf(co2_diff1_seas,  lag.max = 64,main="Figure 11: Seasonal ACF of Combined Diff CO2 series.")
seasonal_pacf(co2_diff1_seas,  lag.max = 64,main="Seasonal PACF of Combined Diff CO2 series.")
par(mfrow=c(1,1))
m2.co2 = Arima(co2.ts,order=c(0,1,0), seasonal=list(order=c(0,1,1), period=12))
res.m2 = residuals(m2.co2);
plot(res.m2,xlab='Time',ylab='Residuals',main=" Figure 12: Time series plot of the residuals")
time_index <- time(res.m2)
trend_model <- lm(res.m2 ~ time_index)
abline(trend_model, col='red', lwd=2)

par(mfrow=c(1,2))
seasonal_acf(res.m2, lag.max = 48, main = "Figure 13: ACF of the residuals")
seasonal_pacf(res.m2, lag.max = 48, main = "PACF of the residuals")
m3.co2 = Arima(co2.ts,order=c(0,2,0), seasonal=list(order=c(0,1,1), period=12))
res.m3 = residuals(m3.co2);
plot(res.m3,xlab='Time',ylab='Residuals',main="Figure 14: Time series plot of the residuals")
time_index <- time(res.m3)
trend_model <- lm(res.m3 ~ time_index)
abline(trend_model, col='red', lwd=2)

par(mfrow=c(1,2))
seasonal_acf(res.m3, lag.max = 48, main = "Figure 15: ACF of the residuals")
seasonal_pacf(res.m3, lag.max = 48, main = "PACF of the residuals")
eacf(res.m3)
#From EACF we have (0,2,1)x(0,1,1), (0,2,2)x(0,1,1), (1,2,2)x(0,1,1)
plot(armasubsets(y=res.m3 , nar=5 , nma=5, y.name='p', ar.method='ols'))
#From BIC table we have (1,2,2), (2,2,2), (3,2,2)
#SARIMA(0,2,1)x(0,1,1)_12
m3_021 = Arima(co2.ts,order=c(0,2,1),seasonal=list(order=c(0,1,1), period=12),method = "ML")
coeftest(m3_021)
```

```r
residual.analysis(model = m3_021)

m3_021.CSS = Arima(co2.ts,order=c(0,2,1),seasonal=list(order=c(0,1,1), period=12),method = "CSS")
coeftest(m3_021.CSS)
residual.analysis(model = m3_021.CSS)
#SARIMA(0,2,2)x(0,1,1)_12

m3_022 = Arima(co2.ts,order=c(0,2,2),seasonal=list(order=c(0,1,1), period=12),method = "ML")
coeftest(m3_022)
residual.analysis(model = m3_022)

m3_022.CSS = Arima(co2.ts,order=c(0,2,2),seasonal=list(order=c(0,1,1), period=12),method = "CSS")
coeftest(m3_022.CSS)
residual.analysis(model = m3_022.CSS)
#SARIMA(1,2,2)x(0,1,1)_12

m3_122 = Arima(co2.ts,order=c(1,2,2),seasonal=list(order=c(0,1,1), period=12),method = "ML")
coeftest(m3_122)
residual.analysis(model = m3_122)

m3_122.CSS = Arima(co2.ts,order=c(1,2,2),seasonal=list(order=c(0,1,1), period=12),method = "CSS")
coeftest(m3_122.CSS)
residual.analysis(model = m3_122.CSS)
#SARIMA(2,2,2)x(0,1,1)_12

m3_222 = Arima(co2.ts,order=c(2,2,2),seasonal=list(order=c(0,1,1), period=12),method = "ML")
coeftest(m3_222)
residual.analysis(model = m3_222)

m3_222.CSS = Arima(co2.ts,order=c(2,2,2),seasonal=list(order=c(0,1,1), period=12),method = "CSS")
coeftest(m3_222.CSS)
residual.analysis(model = m3_222.CSS)

m3_222.CSSML = Arima(co2.ts,order=c(2,2,2),seasonal=list(order=c(0,1,1), period=12),method = "CSS-ML")
coeftest(m3_222.CSSML)
residual.analysis(model = m3_222.CSSML)
m3_321 = Arima(co2.ts,order=c(3,2,1),seasonal=list(order=c(0,1,1), period=12),method = "ML")
coeftest(m3_321)
residual.analysis(model = m3_321)

m3_321.CSS = Arima(co2.ts,order=c(3,2,1),seasonal=list(order=c(0,1,1), period=12),method = "CSS")
coeftest(m3_321.CSS)
residual.analysis(model = m3_321.CSS)

m3_321.CSSML = Arima(co2.ts,order=c(3,2,1),seasonal=list(order=c(0,1,1), period=12),method = "CSS-ML")
coeftest(m3_321.CSSML)
residual.analysis(model = m3_321.CSSML)
m3_322 = Arima(co2.ts,order=c(3,2,2),seasonal=list(order=c(0,1,1), period=12),method = "ML")
coeftest(m3_322)
residual.analysis(model = m3_322)

m3_322.CSS = Arima(co2.ts,order=c(3,2,2),seasonal=list(order=c(0,1,1), period=12),method = "CSS")
coeftest(m3_322.CSS)
```

```r
residual.analysis(model = m3_322.CSS)

m3_322.CSSML = Arima(co2.ts,order=c(3,2,2),seasonal=list(order=c(0,1,1), period=12),method = "CSS-ML")
coeftest(m3_322.CSSML)
residual.analysis(model = m3_322.CSSML)
m3_421 = Arima(co2.ts,order=c(4,2,1),seasonal=list(order=c(0,1,1), period=12),method = "ML")
coeftest(m3_421)
residual.analysis(model = m3_421)

m3_421.CSS = Arima(co2.ts,order=c(4,2,1),seasonal=list(order=c(0,1,1), period=12),method = "CSS")
coeftest(m3_421.CSS)
residual.analysis(model = m3_421.CSS)
m3_422 = Arima(co2.ts,order=c(4,2,2),seasonal=list(order=c(0,1,1), period=12),method = "ML")
coeftest(m3_422)
residual.analysis(model = m3_422)

m3_422.CSS = Arima(co2.ts,order=c(4,2,2),seasonal=list(order=c(0,1,1), period=12),method = "CSS")
coeftest(m3_422.CSS)
residual.analysis(model = m3_422.CSS)
sc.AIC <- AIC(m3_021, m3_022, m3_122, m3_222, m3_321, m3_322, m3_421, m3_422)
sc.BIC <- BIC(m3_021, m3_022, m3_122, m3_222, m3_321, m3_322, m3_421, m3_422)

sort.score(sc.AIC, score = "aic")
sort.score(sc.BIC, score = "bic")
Sm3_021 <- accuracy(m3_021)[1:7]
Sm3_022 <- accuracy(m3_022)[1:7]
Sm3_122 <- accuracy(m3_122)[1:7]
Sm3_222 <- accuracy(m3_222)[1:7]
Sm3_321 <- accuracy(m3_321)[1:7]
Sm3_322 <- accuracy(m3_322)[1:7]
Sm3_421 <- accuracy(m3_421)[1:7]
Sm3_422 <- accuracy(m3_422)[1:7]

df.SARIMA.models <- data.frame(
  rbind(Sm3_021, Sm3_022, Sm3_122, Sm3_222, Sm3_321, Sm3_322, Sm3_421, Sm3_422)
)
colnames(df.SARIMA.models) <- c("ME", "RMSE", "MAE", "MPE", "MAPE", "MASE", "ACF1")
rownames(df.SARIMA.models) <- c("SARIMA(0,2,1)x(0,1,1)", "SARIMA(0,2,2)x(0,1,1)",
                                "SARIMA(1,2,2)x(0,1,1)", "SARIMA(2,2,2)x(0,1,1)",
                                "SARIMA(3,2,1)x(0,1,1)", "SARIMA(3,2,2)x(0,1,1)",
                                "SARIMA(4,2,1)x(0,1,1)", "SARIMA(4,2,2)x(0,1,1)")
round(df.SARIMA.models, 3)
kable(round(df.SARIMA.models, 3), caption = "Forecast Accuracy Metrics for SARIMA Models")
m3_222 = Arima(co2.ts,order=c(2,2,2),seasonal=list(order=c(0,1,1), period=12),method = "ML")
coeftest(m3_222)
residual.analysis(model = m3_222)
#AR(2) is insignificant
m3_123 = Arima(co2.ts,order=c(1,2,3),seasonal=list(order=c(0,1,1), period=12),method = "ML")
coeftest(m3_123)
residual.analysis(model = m3_123)
forecast.co2 = forecast(m3_122, h = 10)
forecast.co2
plot(forecast.co2,
```

```
     main = "Figure 16: CO  Forecast Using SARIMA(1,2,2)(0,1,1)[12]",
     xlab = "Time",
     ylab = "CO  Levels",
     col = "black",
     fcol = "darkgreen",
     flty = 2,
     shadecols = c(rgb(0.6, 0.9, 0.6, 0.3), rgb(0.6, 0.9, 0.6, 0.15)))

# Add fitted values (in blue)
grid(col = "gray80", lty = "dotted")
lines(fitted(forecast.co2), col = "blue", lwd = 2)

# Add legend
legend("topleft",
       legend = c("Observed", "Fitted", "Forecast", "80%/95% Prediction Intervals"),
       col = c("black", "blue", "darkgreen", rgb(0.6, 0.9, 0.6)),
       lty = c(1, 1, 2, NA),
       lwd = c(1, 2, 2, NA),
       pch = c(NA, NA, NA, 15),
       pt.cex = 2,
       bty = "n")
```